Spatial networks and inequalities

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Economic Complexity, Class 7 – Nov 4 2026

Social Data Science, Corvinus University of Budapest


Why do inequalities grow in a small world?
facebook
December 2010
2

Network fragmentation and inequalities

Small-world social networks are fragmented by groups.

One of the strongest forces of group formation is prosperity.

Diffusion of innovation is unequal across network fragments.



Currarini, Pin, Jackson (2009) Econometrica

Today

1. The geography of networks, fragmentation, urban topology / Balázs

2. Social media and urban inequalities / Sándor

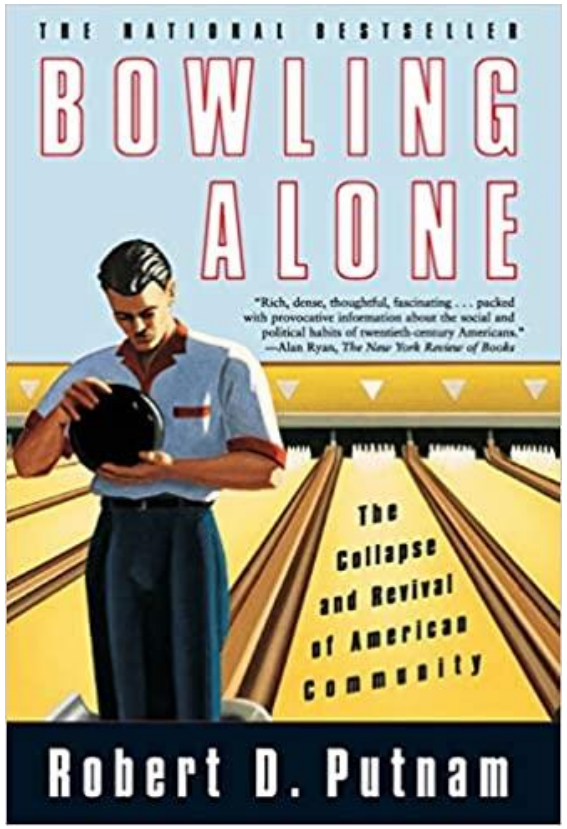
1. The geography of networks, fragmentation, urban topology / Balázs

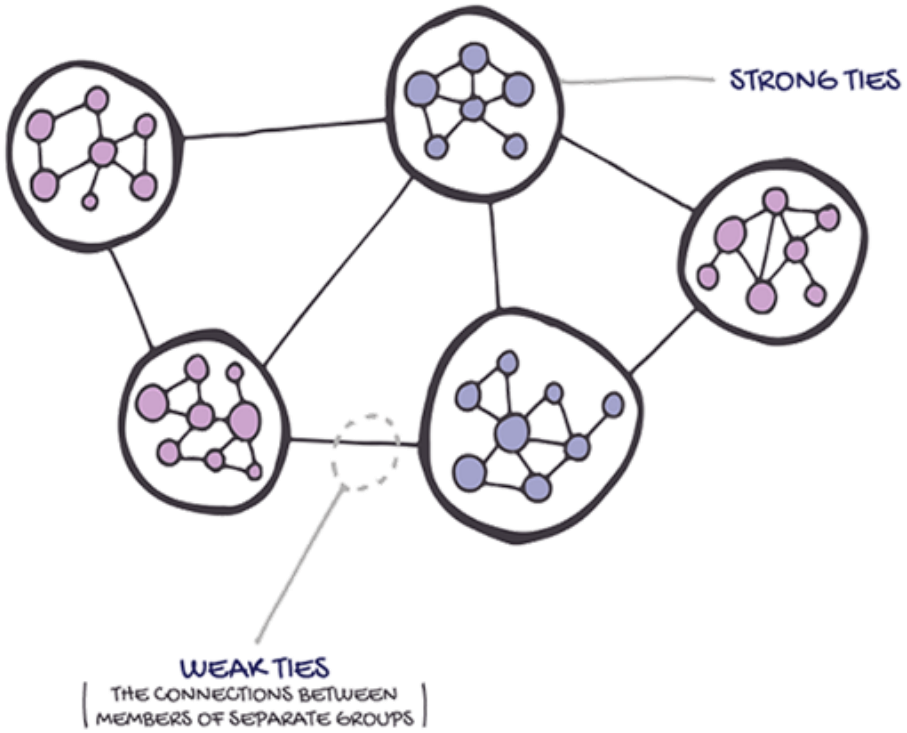
Social capital

The to mobilize resources through social contacts. (Putnam)

Bonding social capital: strong connections (Coleman)

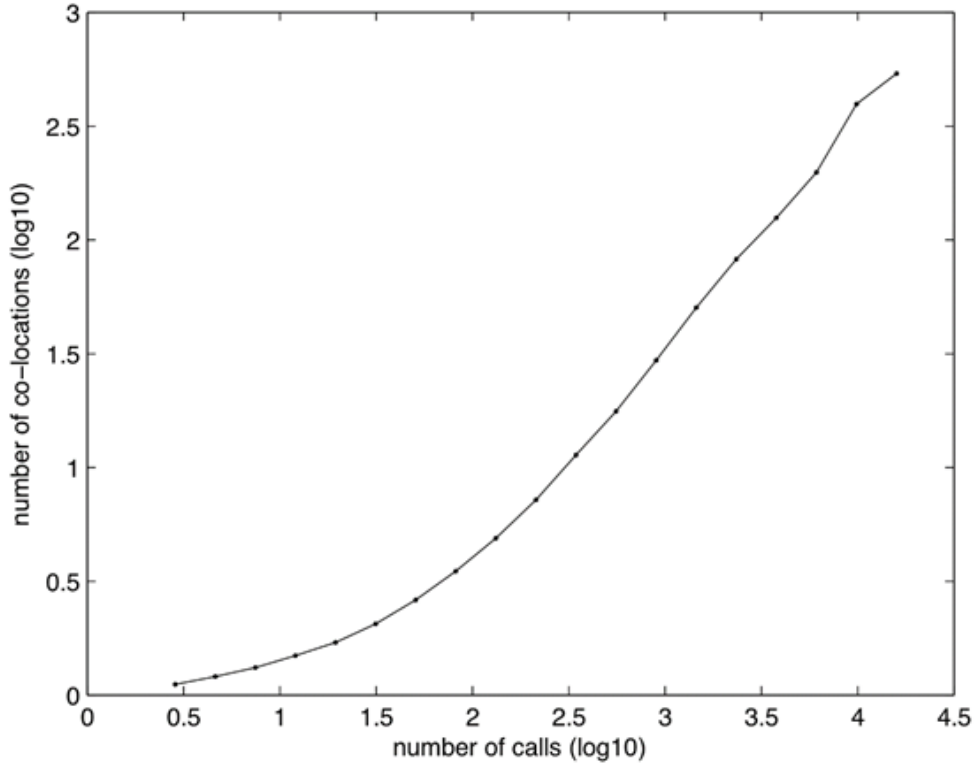
Bridging social capital: weak tie (Granovetter)



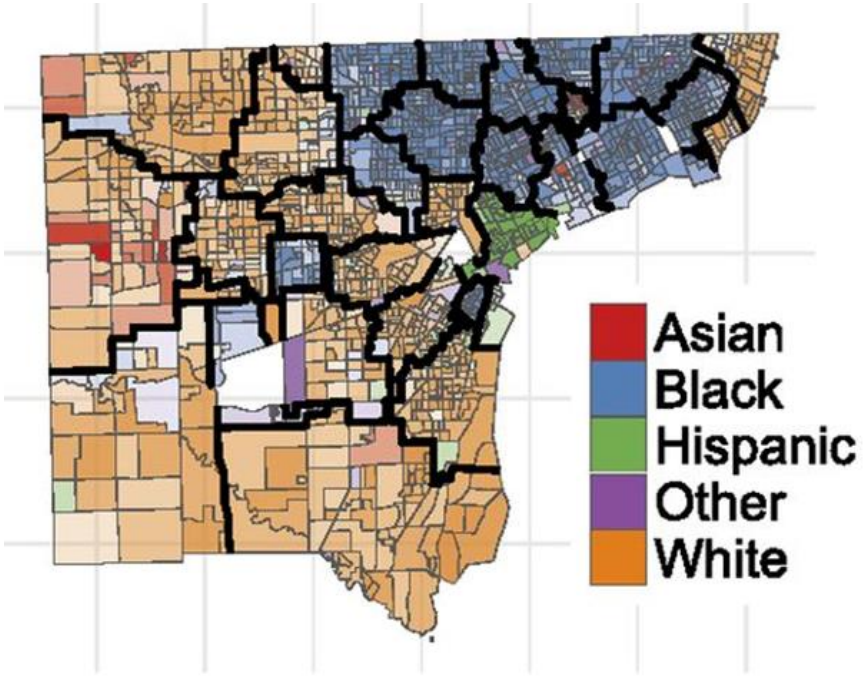


Social capital and social networks in cities

Geographic constraints of social ties increase segregation



Calabrese-Smoreda-Blondel-Ratti (2011) PLOS ONE



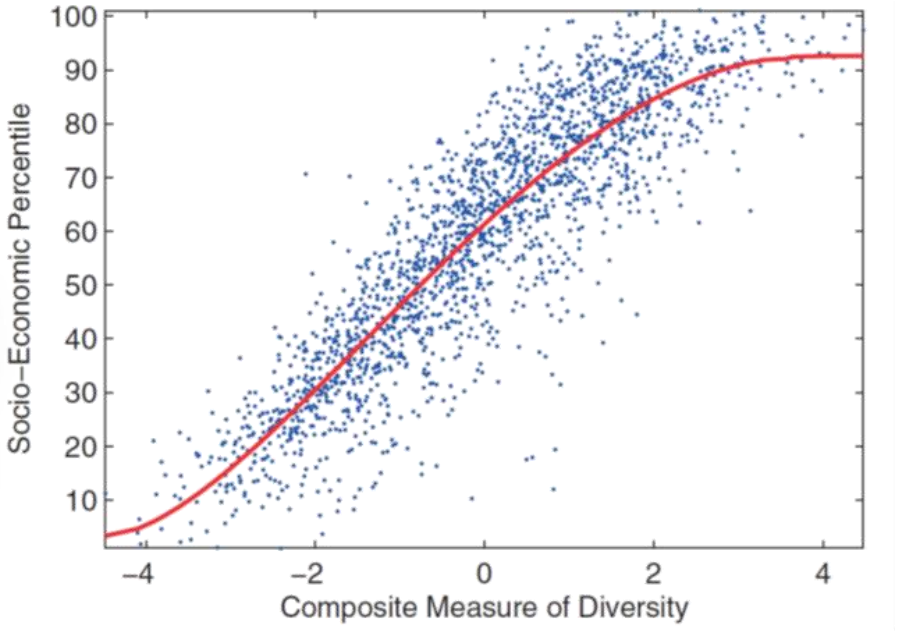
(a) Detroit, $k = 34$

Philip S. Chodrow PNAS 2017;114:44:11591-11596

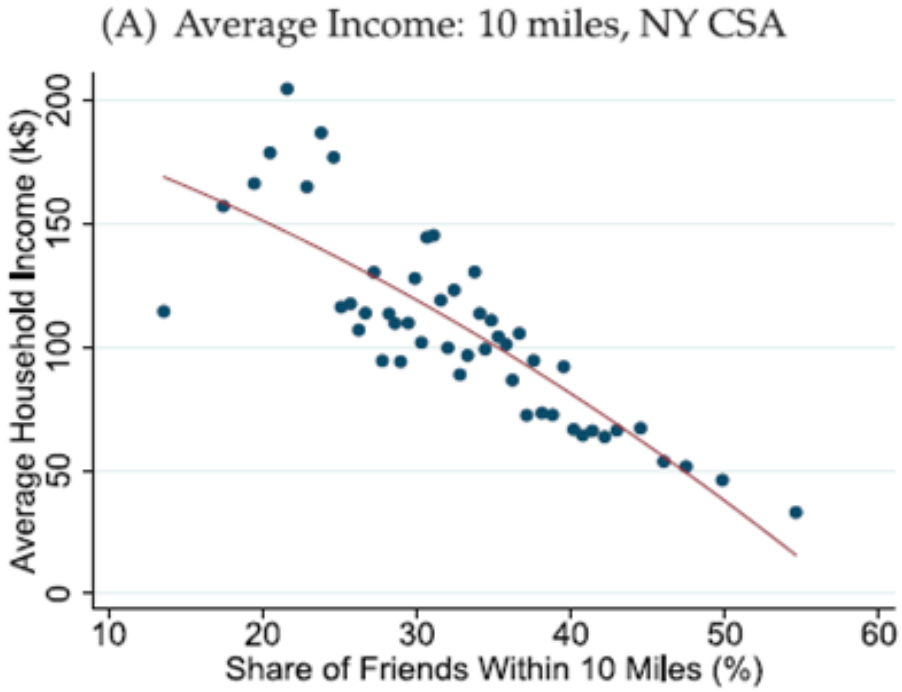
Social capital and social networks in cities

Geographic constraints of social ties predict socio-economic status





Eagle-Macy-Claxton (2010) Science



Bailey et al. (2020) JUE

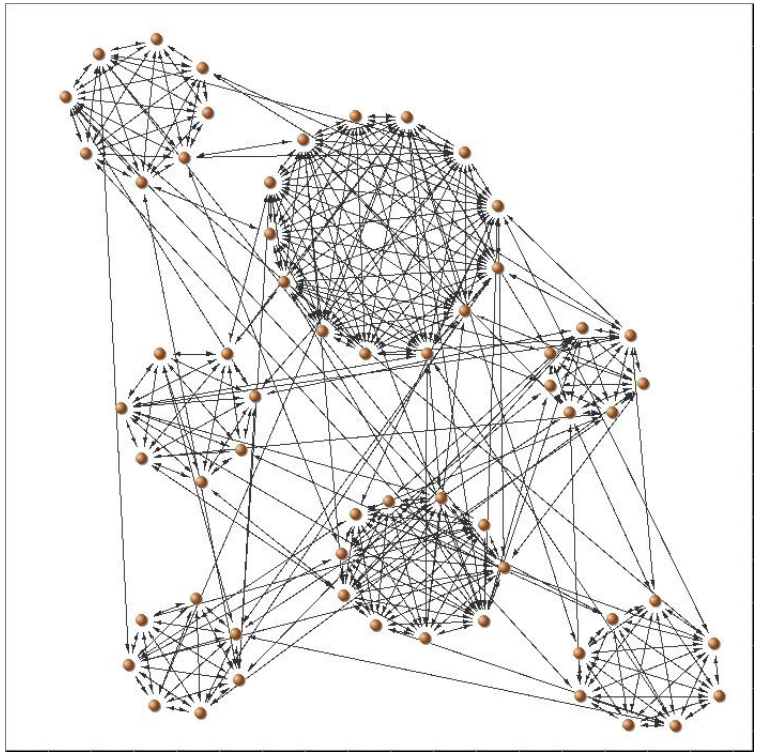
Homophily in small worlds

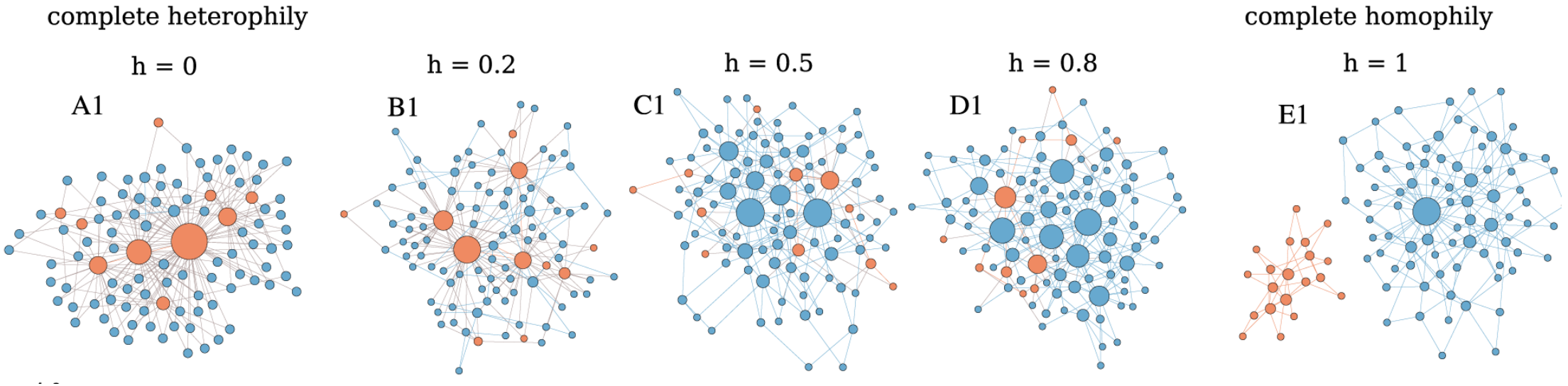
- Social networks cannot be described by random ties:

  Dense local relations and

  few bridges between isolated groups

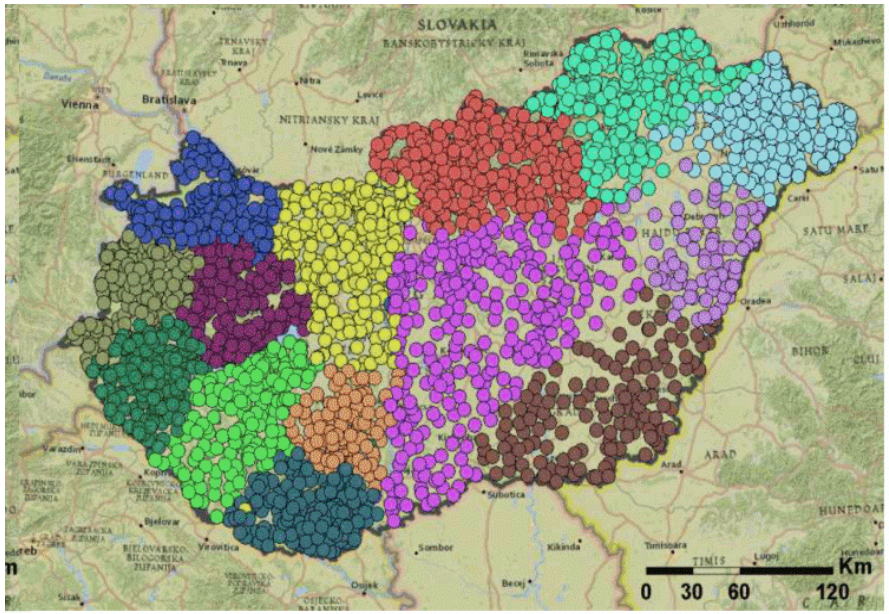
- Small worlds
  - High clustering – due to dense local networks
  - Short paths – due to bridges

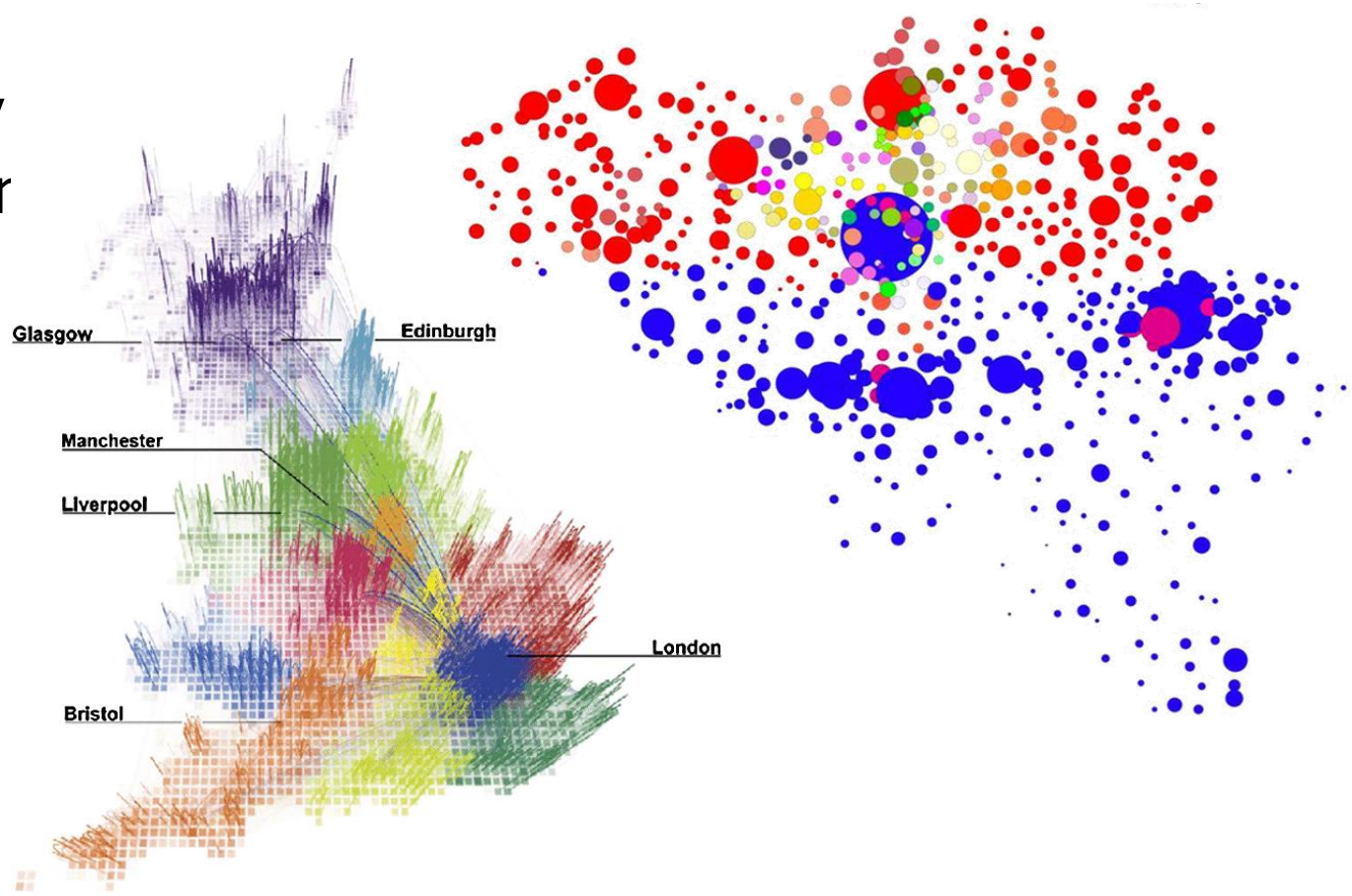




Spatial modules

- Social networks break down by administrative or cultural regior





Expert et al. (2011) PNAS; Sobolevsky et al. (2013) PLoS ONE; Lengyel et al (2015) PLOS ONE

Residential segregation and social networks

Article | Open access | Published: 01 August 2022

Social capital I: measurement and associations with economic mobility

Raj Chetty ✉, Matthew O. Jackson ✉, Theresa Kuchler ✉, Johannes Stroebel ✉, Nathaniel Hendren, Robert B. Fluegge, Sara Gong, Federico Gonzalez, Armelle Grondin, Matthew Jacob, Drew Johnston, Martin Koenen, Eduardo Laguna-Muggenburg, Florian Mudekereza, Tom Rutter, Nicolaj Thor, Wilbur Townsend, Ruby Zhang, Mike Bailey, Pablo Barberá, Monica Bhole & Nils Wernerfelt

Nature **608**, 108–121 (2022) | Cite this article

182k Accesses | **151** Citations | **2296** Altmetric | Metrics

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ARTICLE

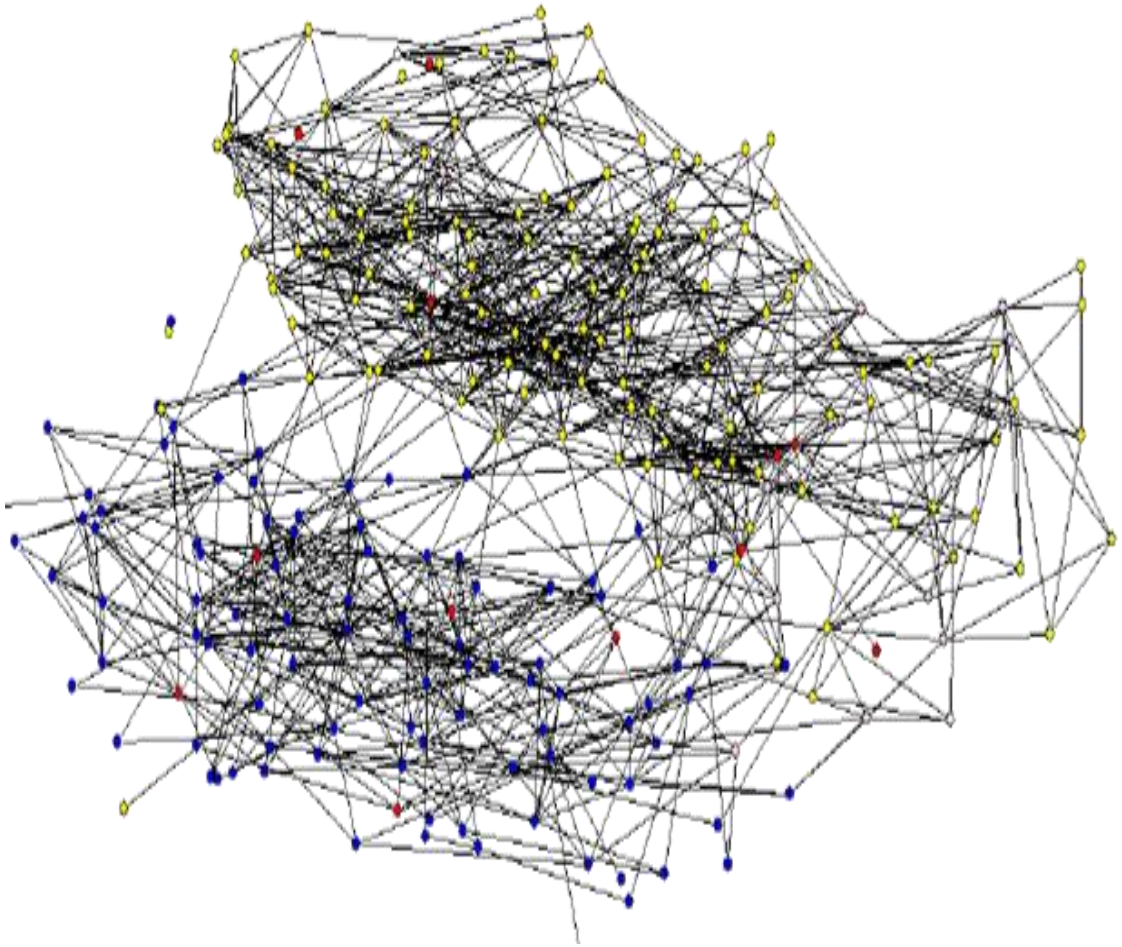
https://doi.org/10.1038/s41467-021-21465-0 OPEN

Inequality is rising where social network segregation interacts with urban topology

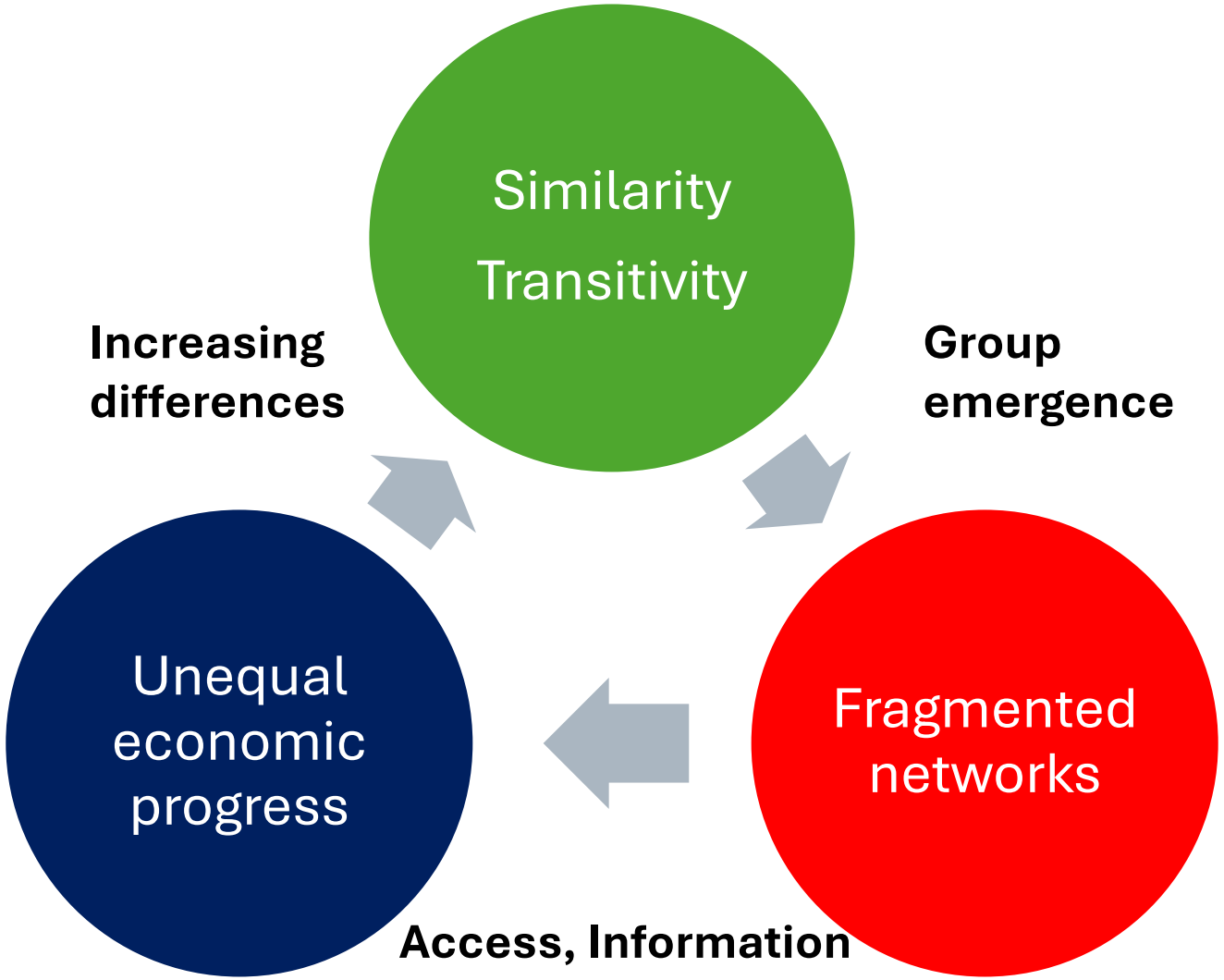
Gergő Tóth[1,2,13], Johannes Wachs[3,4,13], Riccardo Di Clemente[5,6], Ákos Jakobi[7,8], Bence Ságvári[1,9,10], János Kertész[11] & Balázs Lengyel[1,10,12]✉

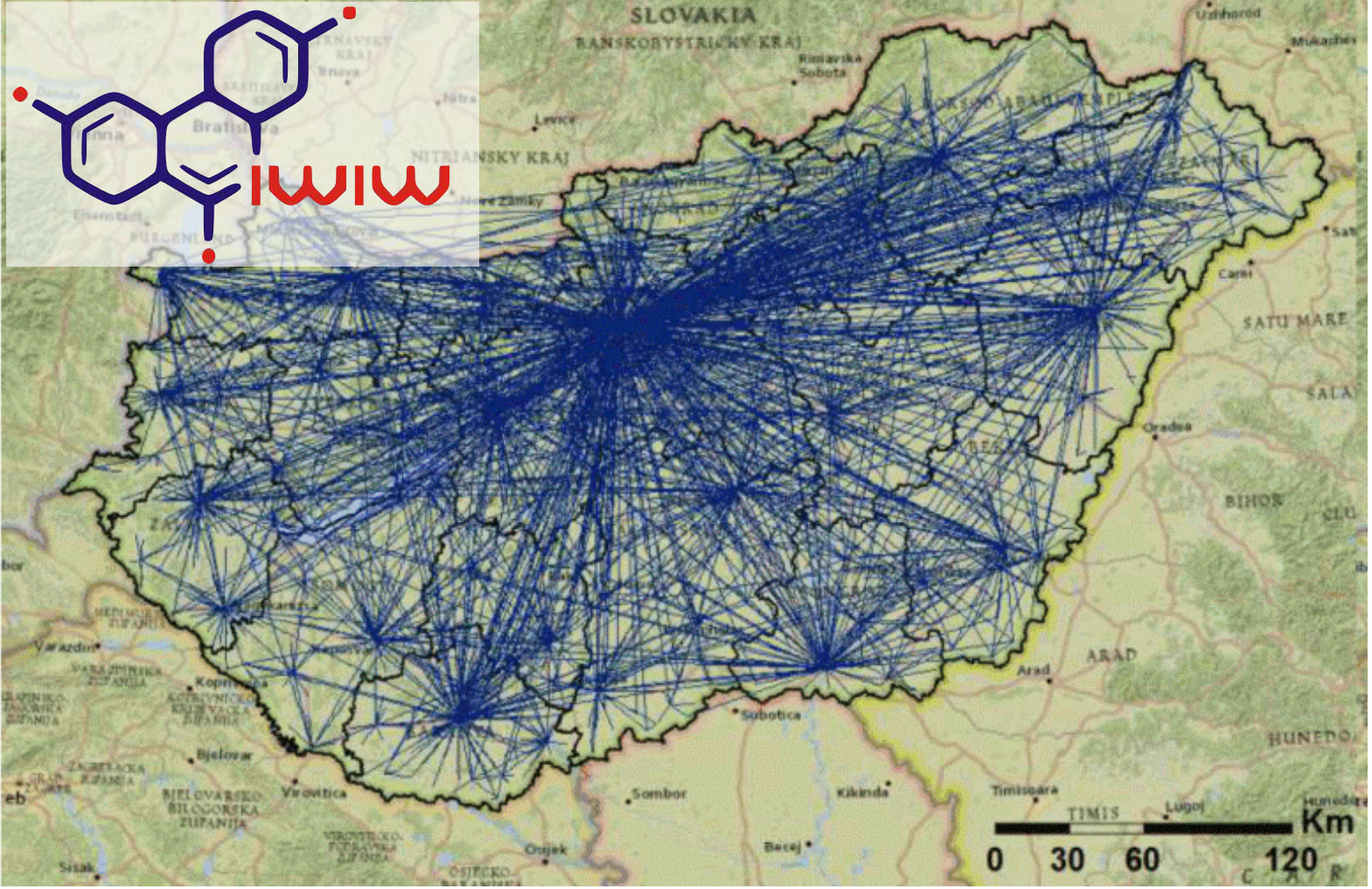
Social networks and inequalities

DiMaggio-Garip (2012) Annual Review of Sociology

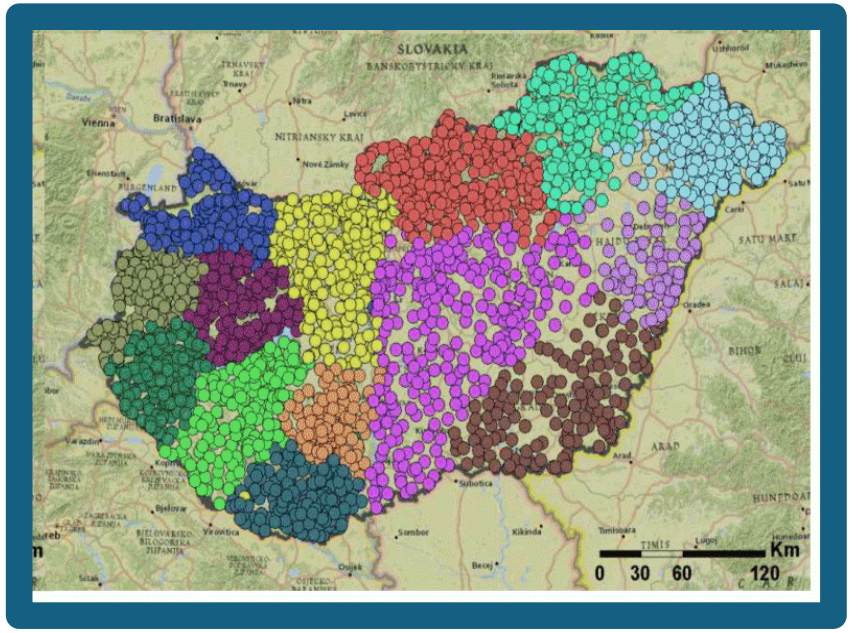


Currarini-Pin-Jackson (2009) Econometrica









2.8 M Users

Social media platform

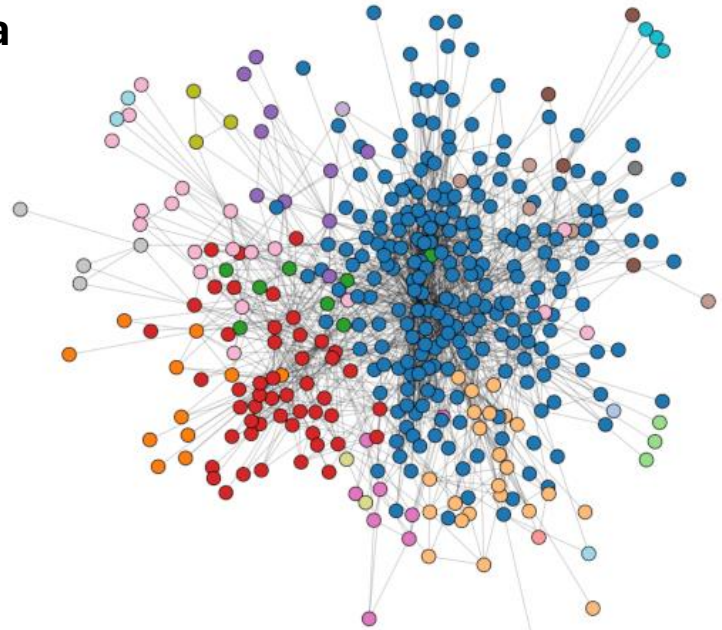
- **Location** (self-reported),
- **Date of registration,**
- **Date of last login,**
- **ID of friends,**
- **ID of invitor**

Jakobi and Lengyel (2014) Springer; Lengyel et al. (2015) PLoS ONE

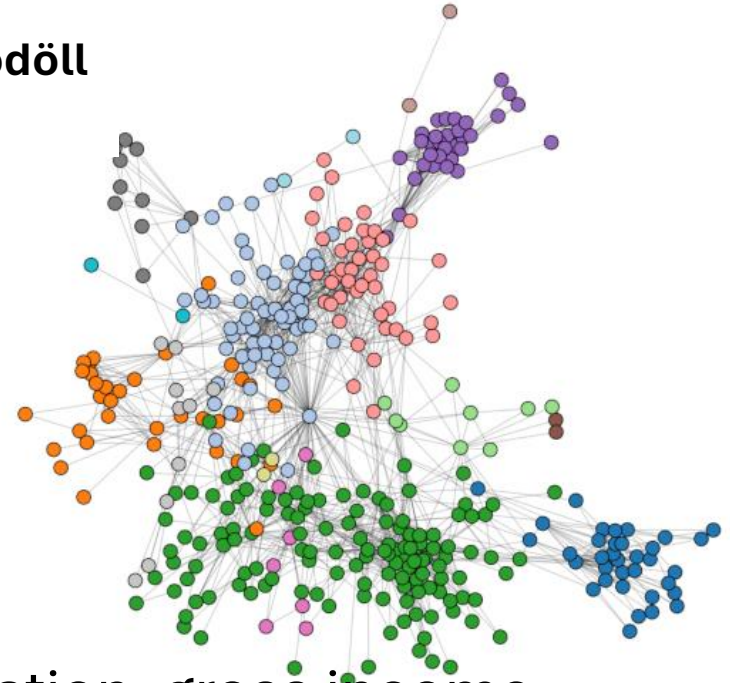
Fragmentation of the network within towns

$$Q_i = \sum_{k=1}^{K_i} \left[\frac{L_k^w}{L_i} - \left(\frac{L_k}{L_i} \right)^2 \right] \qquad F_i = Q_i / Q_i^{max}$$

Ajka

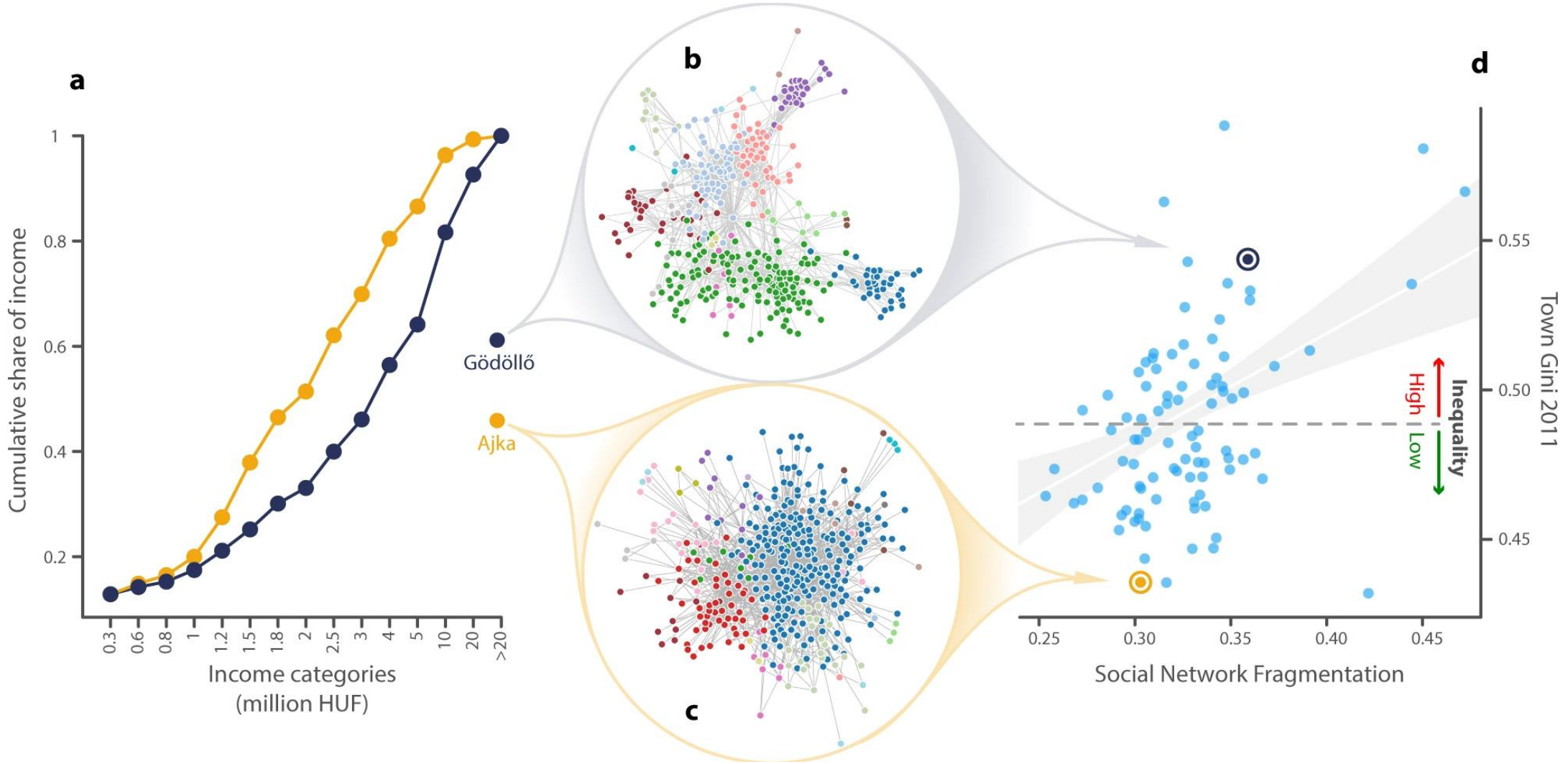


Gödöllő



- TEIR: income groups in towns – population, gross income
- Towns larger than 2500 (except Budapest)
- 2011, 2016

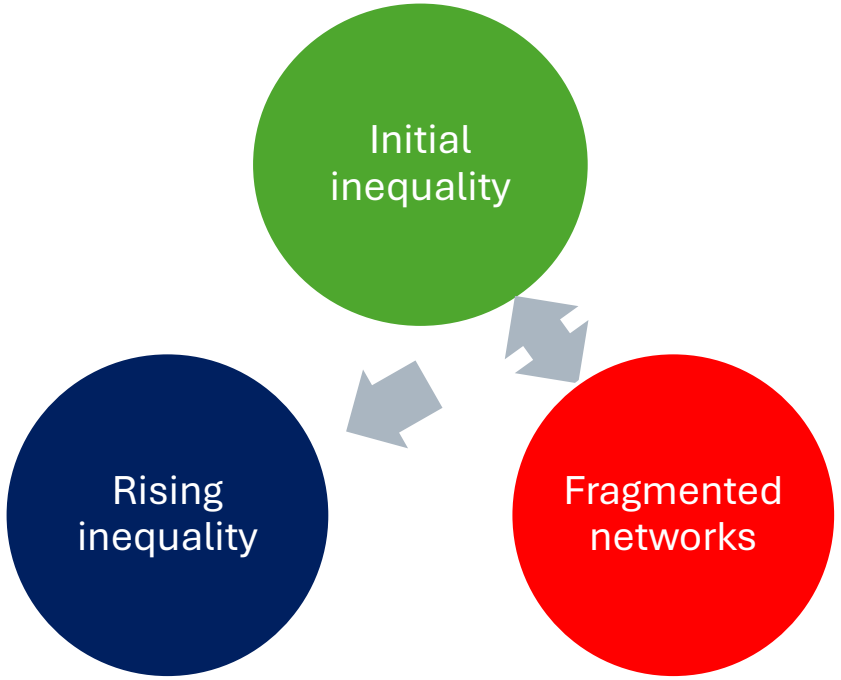
Network fragmentation is correlated with inequalities

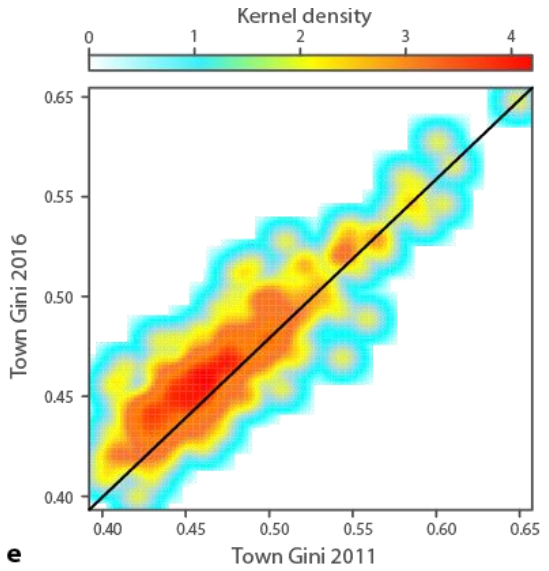


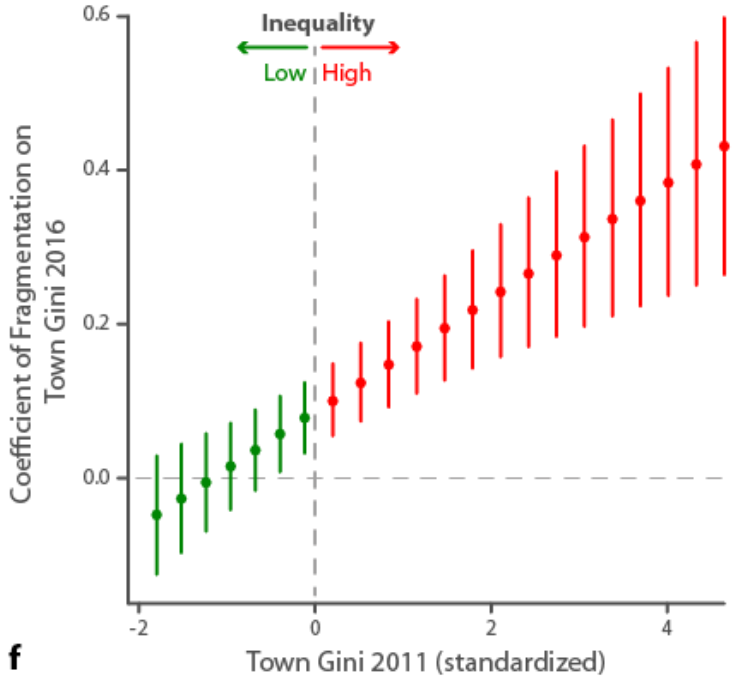
Fragmented networks contribute to existing inequalities

$$G_{i,2016} = \alpha \times G_{i,2011} + \beta \times F_i + \gamma \times (G_{i,2011} \times F_i) + Z_{i,2011} + \epsilon$$

$$\frac{\partial G_{i,2016}}{\partial F_i} = \beta + \gamma \times G_{i,2011}$$





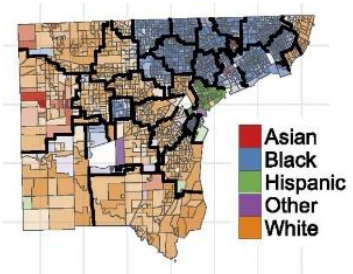


What is the role of urban topology?

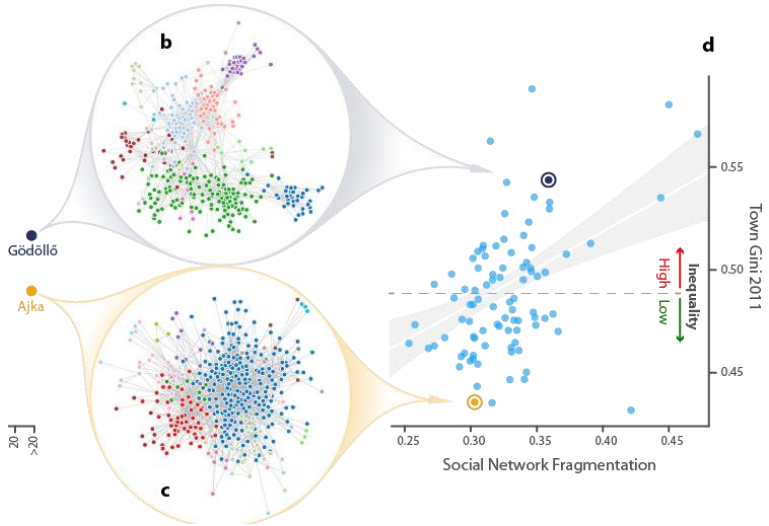
A 2SLS approach

$$F_i = \delta + \gamma IV_i + \delta N_i + e_i$$

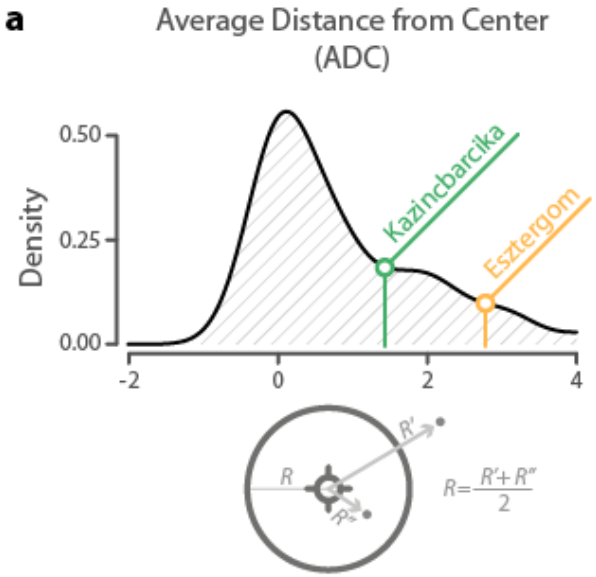
$$G_i = \alpha + \beta_1 \hat{F}_i + \beta_2 X_i + \varphi_k + e_i$$

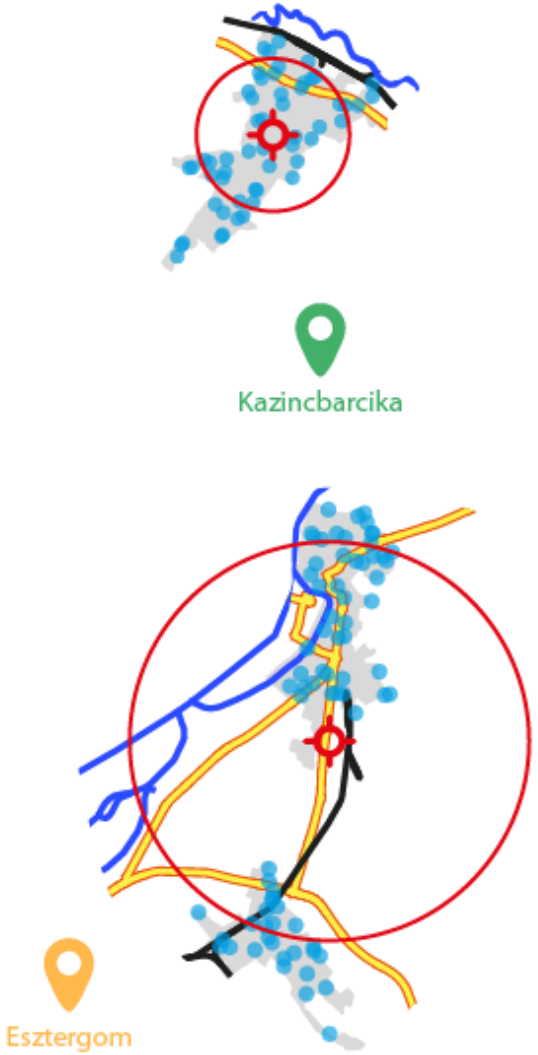


(a) Detroit, $k = 34$



Urban topology 1: average distance from center







Downtown functions as major hub for social interactions
Urban economics standard models (e.g. von Thünen, Alonso)

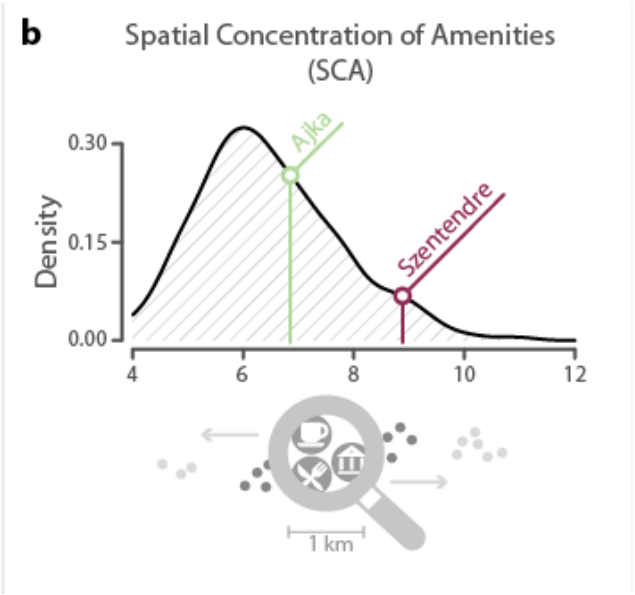
$$ADC_i = \frac{\sum_p^P D_{p,c}}{P} / S_i$$

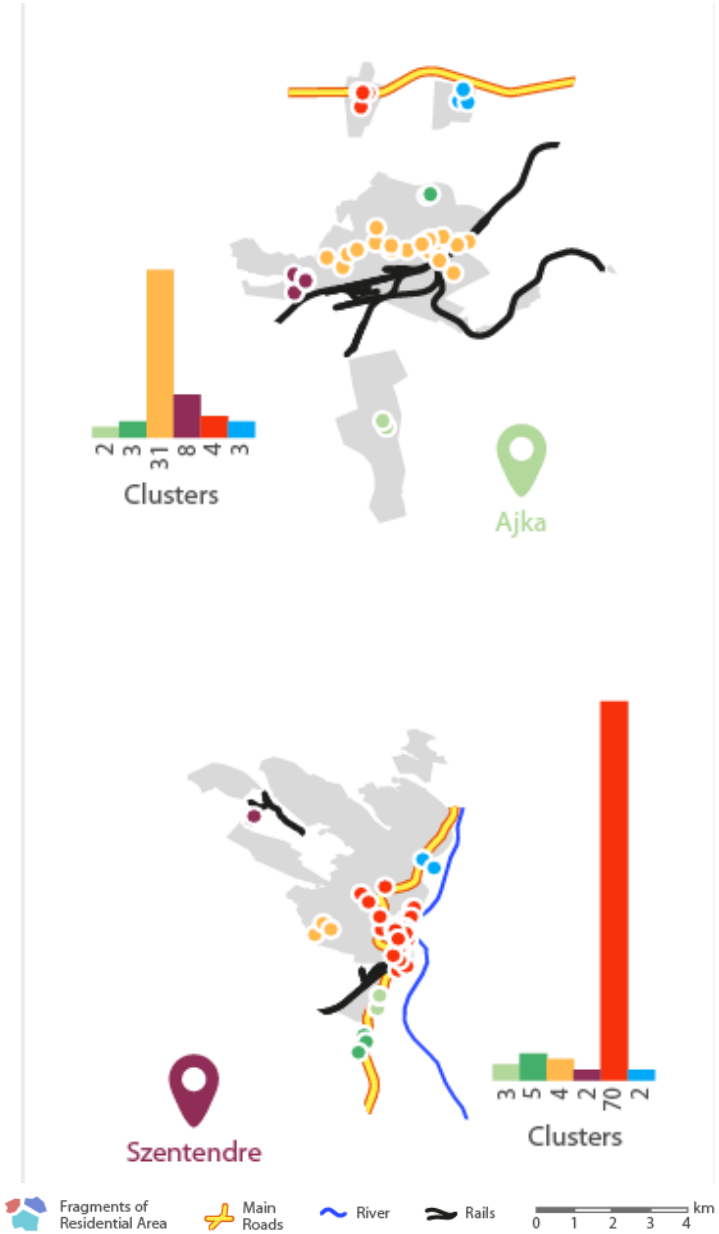
Co-location is important for social interaction and the probability of links decreases as distance grows

-> we expect that fragmentation is higher in towns where ADC is high

Reversed causality is not likely
-> city growth is not likely to be driven by segregated social groups

Urban topology 2: spatial concentration of amenities





H1: spatial concentration of amenities facilitates segregation of the poor living in the periphery
(Brueckner-Thisse-Zenou, 1999, European Economic Review)

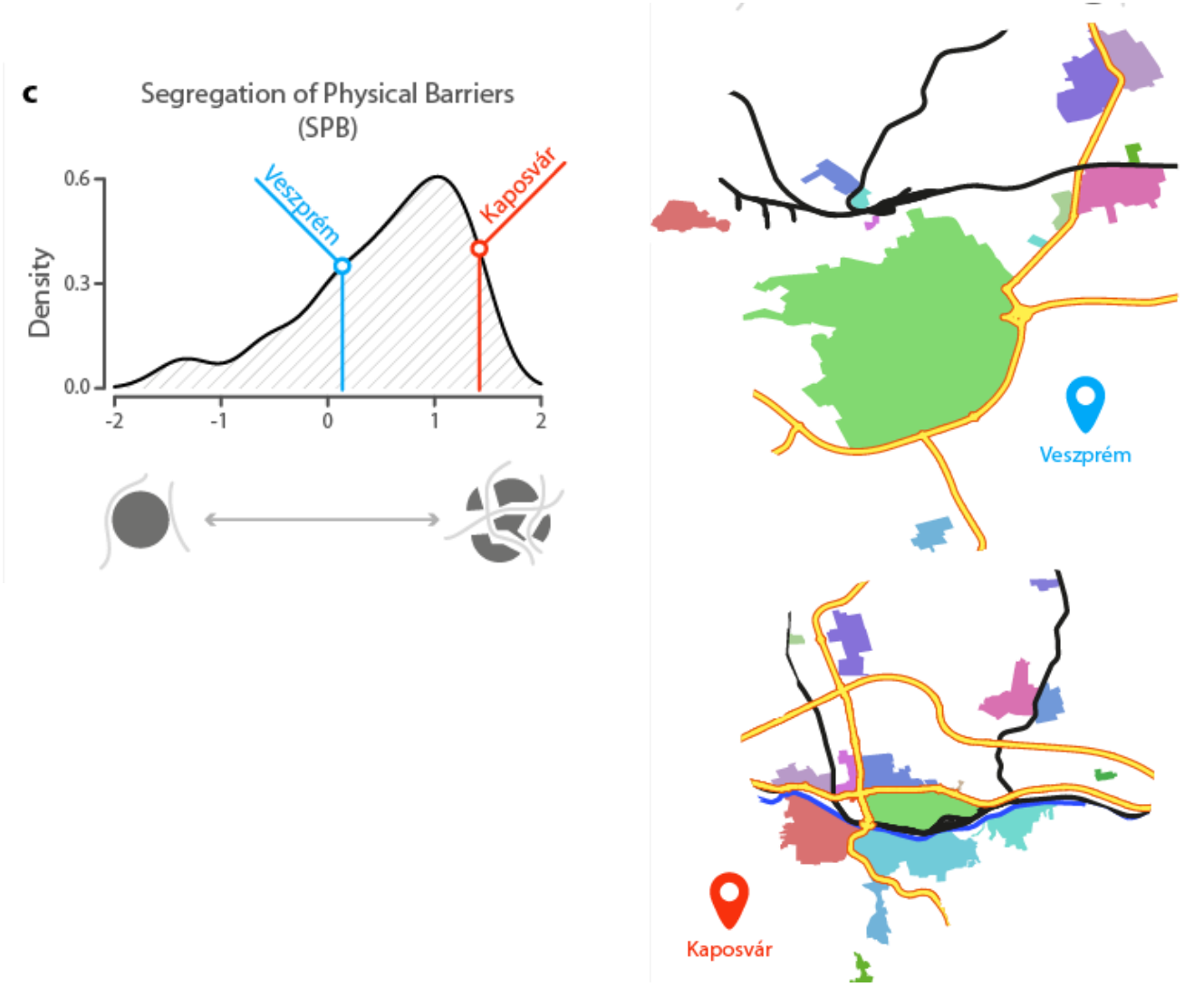
H2: spatial diffuse of amenities provide meeting places in peripheral neighborhoods that facilitates segregation (Zhang-Pryce, 2019, Urban Studies)

$$SCA_i = \frac{\sum_c (p_c \times \log p_c)}{n(c)} / S_i$$

H1 is verified with a positive correlation between the measure and fragmentation.

-> we expect that fragmentation is higher in towns where SCA is high even when controlled for town characteristics.

Urban topology 3: segregation of physical barriers



Built and natural barriers facilitate segregation in cities (Cutler and Glaeser, 1997, QJE)

Used to instrument racial segregation effect on inequality (Ananat, 2011, AEJ: Applied Economics)

$$SPB_i = 1 - \sum_a (S_a/S_i)^2$$

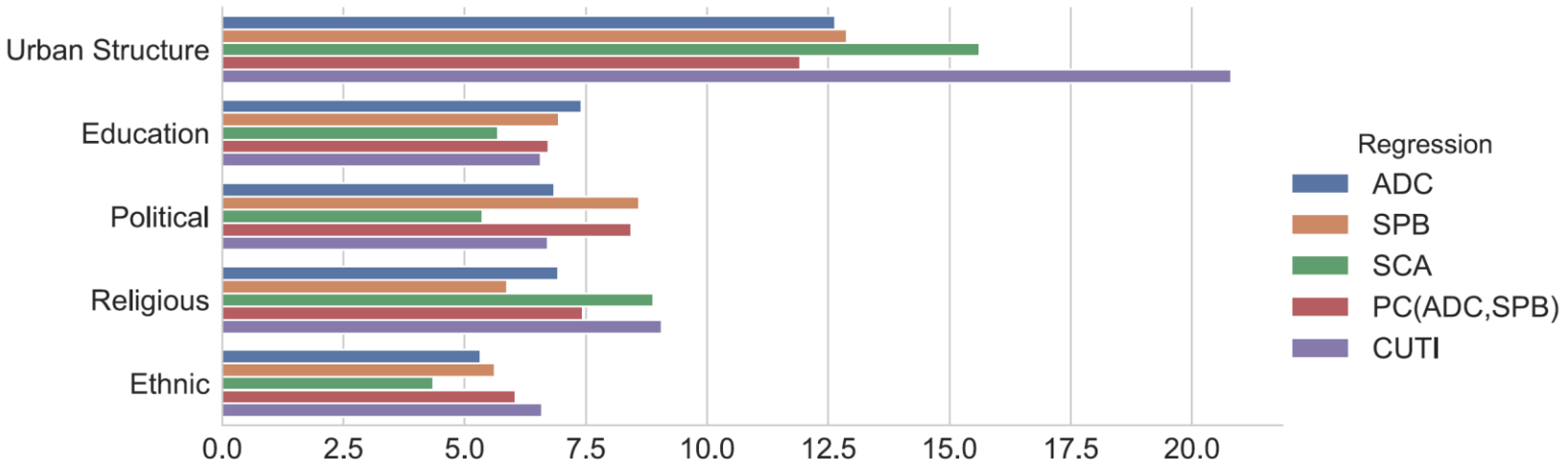
Barriers decrease the probability of face-to-face interaction (Lengye et al, 2015 PLOS ONE)
Barriers facilitate sorting into neighborhoods (Benton, 2017, Journal of Urban History)
-> we expect that fragmentation is higher in towns where SPB is high

Reversed causality is not likely
-> railroads were not built after the 1940s in Hungary (Körner-Nagy, 2011, Építés-Építészettudomány)

Does urban topology explain network segregation better than other dimensions of social segregation?

Random Forest approach:

- **500 regressions randomly combining the explanatory variables and calculate mean squared error (MSE) from the random sample of these regressions**
- **Randomly vary variables around their mean and recalculate MSE**



1. School inequalities

Coefficient of variance in 6th grade math exam.

2. Political fragmentation

Coefficient of variance in the votes given to Fidesz across voting districts.

3. Religious fragmentation

Entropy of religious distribution.

4. Ethnic fragmentation

Entropy of ethnic distribution.

Does urban topology correlate with network fragmentation? - 1st stage in the 2SLS analysis

$$F_i = \delta + \gamma IV_i + \delta N_i + e_i$$

	Dependent variable:$Fragmentation(F_i)$		
	(1)	(2)	(3)
ADC	0.091**		
	(0.045)		
SCA		0.110**	
		(0.046)	
SPB			0.168***
			(0.044)
User rate	0.367***	0.355***	0.344***
	(0.045)	(0.046)	(0.044)
Constant	−0.000	−0.000	−0.000
	(0.042)	(0.042)	(0.042)
Observations	473	473	473
R^2	0.167	0.170	0.185
Adjusted R^2	0.163	0.166	0.181
Residual Std. Error (df = 470)	0.915	0.913	0.905
F Statistic (df = 2; 470)	47.082***	48.085***	53.259***
Note:			$^*p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

Urban topology <-> network fragmentation <-> inequality
2nd stage in the 2SLS analysis

$$G_i = \alpha + \beta_1 \hat{F}_i + \beta_2 X_i + \varphi_k + e_i$$

Control variables:
- Level of FDI
- Change of FDI
- Unemployment rate
- Population density
- Distance to the closest border

	Dependent variable: Gini Coefficient ($G_{i,2016}$)		
	Instrumental Variable		
	ADC	*SCA*	*SPB*
Estimated Fragmentation ($\hat{F}_i$)	0.408***	0.533***	0.288**
	(0.153)	(0.146)	(0.138)
Population density	−0.092*	−0.118**	−0.067
	(0.055)	(0.052)	(0.053)
Distance to border	−0.243***	−0.231***	−0.254***
	(0.059)	(0.062)	(0.058)
Constant	−0.386	−0.392	−0.380
	(0.369)	(0.330)	(0.372)
County FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
First Stage F-test	22.290***	24.009***	26.754***
Wu-Hausman test	1.107	3.729	0.011
Sargan test	0.051	5.349*	1.400
Observations	473	473	473
R^2	0.231	0.192	0.245
Adjusted R^2	0.186	0.145	0.200
Res.St.Err. (df = 446)	0.902	0.924	0.894
Note:	*p<0.1; **p<0.05; ***p<0.01		

2. Social media and urban inequalities

Sándor Juhász

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Income-related spatial concentration of individual social capital in cities

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Abstract

Social connections that span across diverse urban neighborhoods can support prosperity by mobilizing social capital. However, there is limited evidence on the spatial structure of individual social capital inside cities. This paper demonstrates that social capital measured by online social connections is spatially more concentrated for residents of lower-income neighborhoods than for residents of higher-income neighborhoods. We map the micro-geography of individual online social networks in the 50 largest metropolitan areas of the United States using a large-scale geolocalized Twitter dataset. We analyze the spatial dimension of individual social capital by the share of friends, closed triangles, and share of supported ties within circles of short distance radii (1, 3, 5, and 10 km) around users' home location. We compare residents from below-median income neighborhoods with above-median income neighborhoods, and find that users living in relatively poorer neighborhoods have a significantly higher share of connections in close proximity. Moreover, their network is more cohesive and supported within a short distance from their home. These patterns prevail across the 50 largest US metropolitan areas with only a few exceptions. The found disparities in the micro-geographic concentration of social capital can feed segregation and income inequality within cities constraining social circles of low-income residents.

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Email: kovacs.adam@krtk.hu

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Universal patterns of long-distance commuting and social assortativity in cities

Eszter Bokányi[1,2]✉, Sándor Juhász[1,2], Márton Karsai[3,4] & Balázs Lengyel[1,2]

Millions commute to work every day in cities and interact with colleagues, partners, friends, and strangers. Commuting facilitates the mixing of people from distant and diverse neighborhoods, but whether this has an imprint on social inclusion or instead, connections remain assortative is less explored. In this paper, we aim to better understand income sorting in social networks inside cities and investigate how commuting distance conditions the online social ties of Twitter users in the 50 largest metropolitan areas of the United States. An above-median commuting distance in cities is linked to more diverse individual networks, moreover, we find that longer commutes are associated with a nearly uniform, moderate reduction of overall social tie assortativity across all cities. This suggests a universal relation between long-distance commutes and the integration of social networks. Our results inform policy that facilitating access across distant neighborhoods can advance the social inclusion of low-income groups.

Cities are champions of diversity[1–3]. Complex interaction networks of individuals in urban areas enabled by population density, co-location, and easy access together made cities the global engines of technological and economic progress[4–7]. However, cities are also known for high levels of segregation[8–10] where disparate neighborhoods are separated from each other in the urban space[11–15]. Furthermore, spatial segregation by income also fragments social networks, which can hinder progress and can deepen inequalities[16–20]. Given the importance of this problem, a growing community has investigated the patterns of mobility in cities to better understand mixing potentials across disparate and diverse neighborhoods[21–24], which may increase economic prosperity[25]. Yet, less is known whether mobility mixing has any imprint on the social connections of people.

Commuting covers a large share of urban mobility[26] and by connecting home with work locations, the places where people spend most of their time, it plays an important role in the spatial formation of social connections[27–29]. Since aggregated social networks form spatially bounded communities across neighborhoods[17], the further one commutes, the higher the likelihood that commuting-related social connections will introduce diversity in the egocentric network of the commuter[30,31]. Due to spatial segregation, economically disparate neighborhoods tend to be far from each other[32], thus long commutes are more likely to link places with different social status[33,34]. Nevertheless, it is not trivial that long commutes should facilitate social inclusion, because social interactions might remain assortative even at places far from home[21,23,35]. Meanwhile, the time to develop new social connections is especially limited for low-income workers who travel to work during rush hours[36,37].

The spatial distribution of high versus low-income households determines the length of travel that can bridge disparate neighborhoods. Since the scale of socio-economic isolation greatly varies across cities[17], one may expect that the mobility of people also enables a different degree of social mixture. However, the assortativity of urban mobility is a universal feature across cities: individuals have been recently reported to visit locations that are similar to their home neighborhood[21,23,38–40]. Yet, how assortativity of commuting and social networks are related and how this relation is modified by the length of commute in cities is still largely uncovered.

In this paper, we aim to better understand how mixing in urban social networks is facilitated by commuting. To answer this question, we use a unique dataset on 348,850 Twitter users living in the 50 largest metropolitan areas of the US and track their home and work locations as well as their mutual followership ties on the platform, which from now on, we call the social network of users. We project these social networks in the urban space and attribute users with an average income based on their home locations on an income map extracted from census

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PNAS RESEARCH ARTICLE | SOCIAL SCIENCES

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Urban highways are barriers to social ties

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Affiliations are included on p. 9.

Edited by Susan Hanson, Clark University, Worcester, MA; received May 7, 2024; accepted January 23, 2025

Urban highways are common, especially in the United States, making cities more car-centric. They promise the annihilation of distance but obstruct pedestrian mobility, thus playing a key role in limiting social interactions locally. Although this limiting role is widely acknowledged in urban studies, the quantitative relationship between urban highways and social ties is barely tested. Here, we define a Barrier Score that relates massive, geolocated online social network data to highways in the 50 largest US cities. At the granularity of individual social ties, we show that urban highways are associated with decreased social connectivity. This barrier effect is especially strong for short distances and consistent with historical cases of highways that were built to purposefully disrupt or isolate Black neighborhoods. By combining spatial infrastructure with social tie data, our method adds a dimension to demographic studies of social segregation. Our study can inform reparative planning for an evidence-based reduction of spatial inequality, and more generally, support a better integration of the social fabric in urban planning.

social network | segregation | highways | barrier effect

Cities are hubs of concentrated social capital that can foster diversity and innovation (1, 2). However, this potential is threatened by spatial fragmentation through built infrastructure that can divide neighborhoods (3, 4), exacerbate inequalities (5, 6), and contribute to segregation (7). Among various types of barriers fragmenting urban areas, roads designed for motorized traffic are the most ubiquitous, especially highways (4, 8, 9). Since the 1960s, urban planners have theorized that high-traffic roads reduce opportunities for creating and maintaining social ties across divided neighborhoods (10), thus undermining the social cohesion essential for the development of thriving communities. This premise lies at the core of contemporary urban planning research and interventions (11, 12) that strive to meet the UN's sustainable development goal of "making cities and human settlements inclusive, safe, resilient, and sustainable" (13).

Despite its significance in urban planning theories, the association between high-traffic roads and reduced social connectivity has never been measured empirically, with the notable exception of a few small-scale, survey-based studies (14, 15). Previous quantitative research in this area, constrained by the scarcity of georeferenced social network data (16, 17), has focused instead on measuring socioeconomic segregation in cities. This goal has been achieved either by using static demographic data (7, 18) or, more recently, through mobility data (19–21), with only sporadic attempts to link segregation to urban barriers (6, 22, 23). While highly valuable, such previous research could not explicitly consider social ties. However, providing an explicit, quantitative estimation of the barrier effect of different roads in curbing social ties is crucial for guiding evidence-based plans of restorative urban interventions and for prioritizing them according to their estimated benefits (24).

To fill this gap, we introduce a method to systematically quantify the association between highways and social ties at multiple scales, ranging from individual highway segments to entire metropolitan areas. We focus on the network of urban highways in the United States. This highway network offers a compelling subject for the study of barrier effects: With a cost of at least 1.4 trillion USD (25), US highways were built to bridge city centers and newly created suburbs; simultaneously, they displaced an estimated 1 million people from their neighborhoods and today pose hard-to-cross physical barriers to pedestrians and cyclists (4, 26).

Onto this network of urban highways within the 50 largest metropolitan areas in the United States, we overlay a massive geolocated social network of ties between individuals who follow each other on Twitter (27). We compute a Barrier Score which quantifies the reduction in the number of social ties crossing highways, comparing the empirical crossings with a null model that makes ties oblivious to highways. The distribution

Significance

Highways are physical barriers that cut opportunities for social connections, but the magnitude of this effect has not been quantified. Such quantitative evidence would enable policy-makers to prioritize interventions that reconnect urban communities—an urgent need in many US cities. We relate urban highways in the 50 largest US cities with massive, geolocated online social network data to quantify the decrease in social connectivity associated with highways. We find that this barrier effect is strong in all 50 cities, and particularly prominent over shorter distances. We also confirm this effect for highways that are historically associated with racial segregation. Our research demonstrates with high granularity the long-lasting impact of decades-old infrastructure on society and provides tools for evidence-based remedies.

Author contributions: L.M.A., A.V., S.J., M.S., and E.B. designed research; L.M.A., A.V., S.J., M.S., and E.B. performed research; L.M.A., A.V., and S.J. contributed new reagents/analytic tools; L.M.A., A.V., S.J., M.S., and E.B. analyzed data; and L.M.A., A.V., S.J., M.S., and E.B. wrote the paper.

The authors declare no competing interest.

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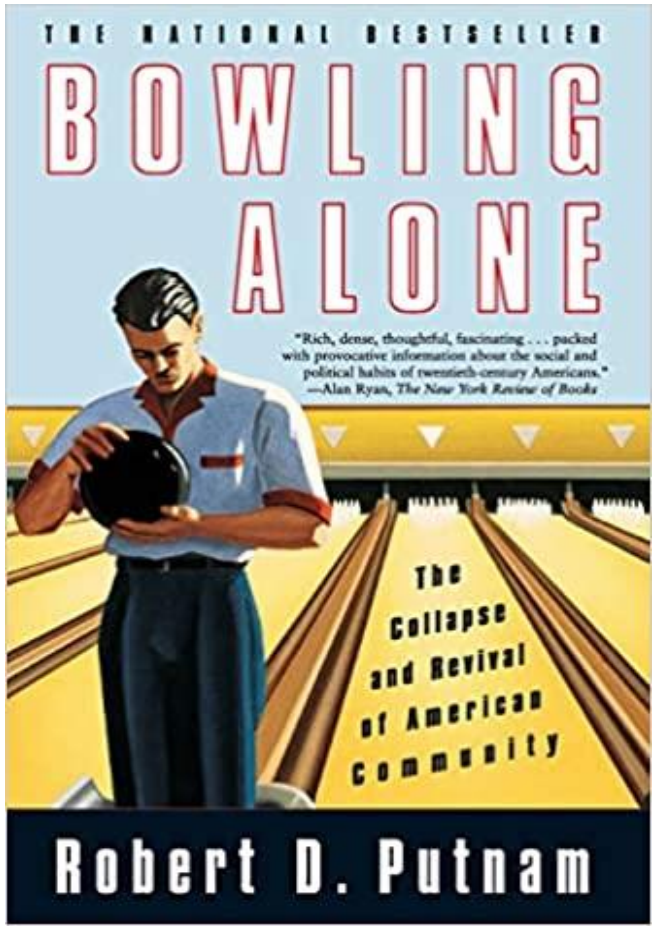
[1]To whom correspondence may be addressed. Email: luai@itu.dk.

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Published March 4, 2025.

3.1) Income-related spatial concentration of individual social capital in cities

Social capital, social networks and geographic constraints



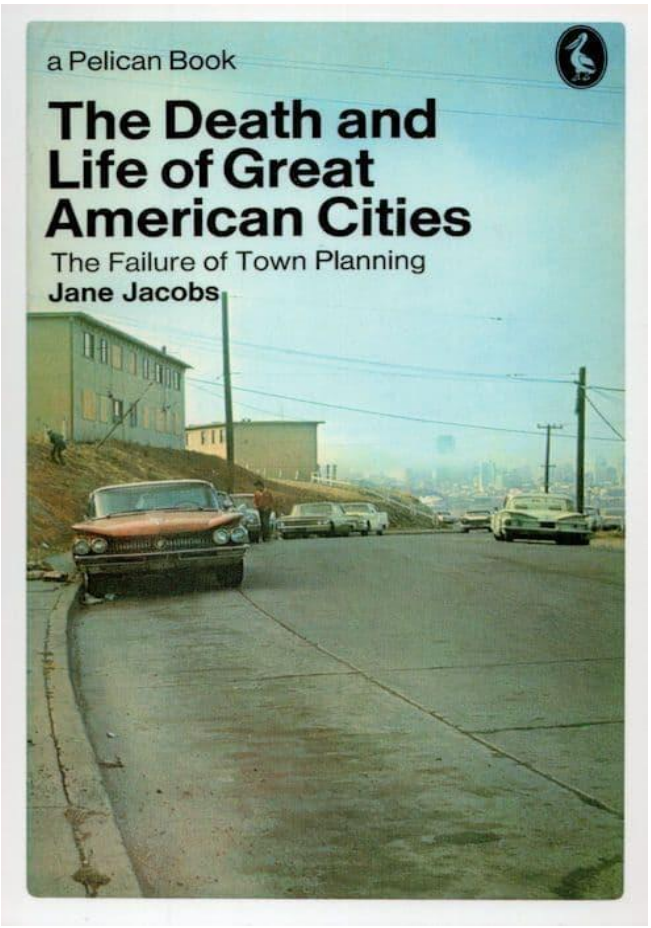
Social Capital in the Creation of Human Capital[1]

James S. Coleman
University of Chicago

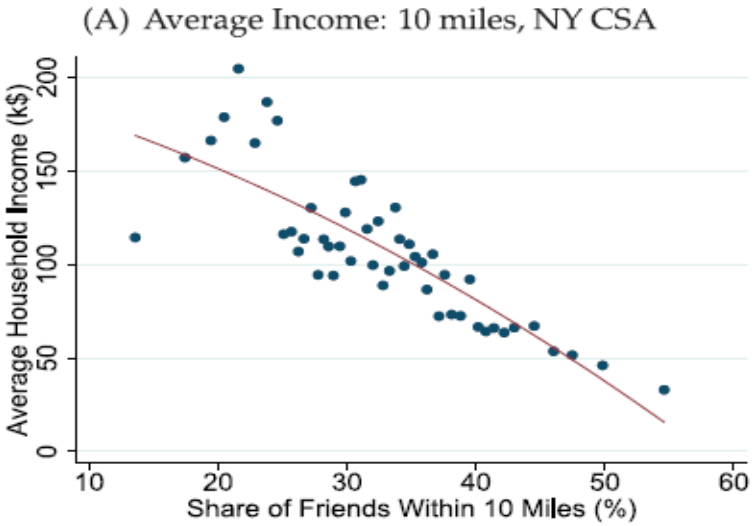
In this paper, the concept of social capital is introduced and illustrated, its forms are described, the social structural conditions under which it arises are examined, and it is used in an analysis of dropouts from high school. Use of the concept of social capital is part of a general theoretical strategy discussed in the paper: taking rational action as a starting point but rejecting the extreme individualistic premises that often accompany it. The conception of social capital as a resource for action is one way of introducing social structure into the rational action paradigm. Three forms of social capital are examined: obligations and expectations, information channels, and social norms. The role of closure in the social structure in facilitating the first and third of these forms of social capital is described. An analysis of the effect of the lack of social capital available to high school sophomores on dropping out of school before graduation is carried out. The effect of social capital within the family and in the community outside the family is examined.

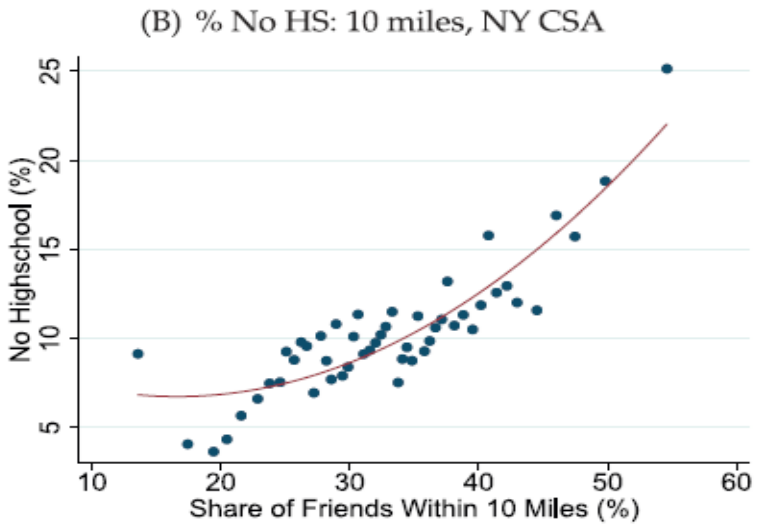
There are two broad intellectual streams in the description and explanation of social action. One, characteristic of the work of most sociologists, sees the actor as socialized and action as governed by social norms, rules, and obligations. The principal virtues of this intellectual stream lie in its ability to describe action in social context and to explain the way action is shaped, constrained, and redirected by the social context.

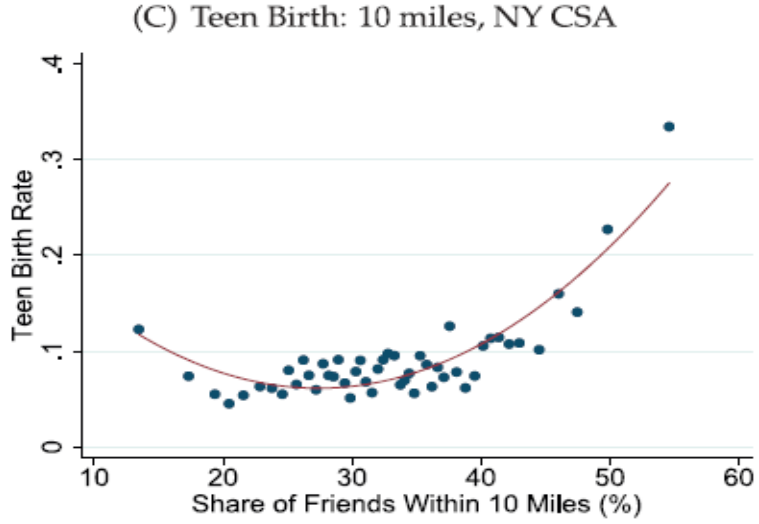
The other intellectual stream, characteristic of the work of most economists, sees the actor as having goals independently arrived at, as acting independently, and as wholly self-interested. Its principal virtue lies in

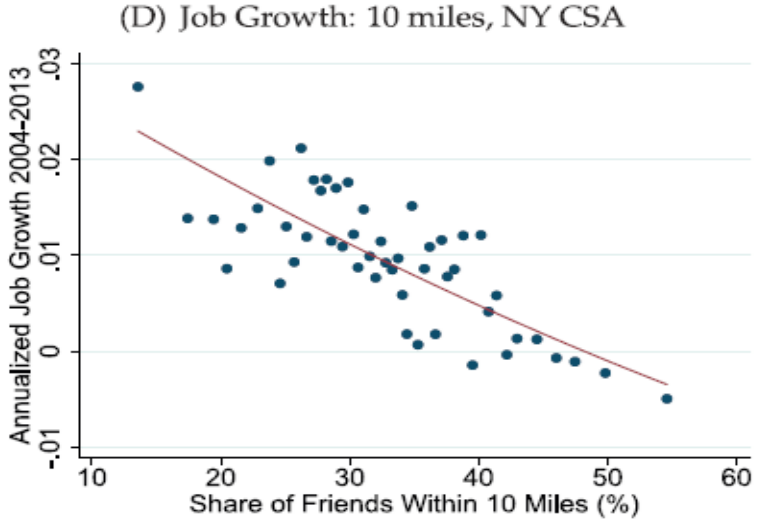


Space and socio-economic dimensions of social ties









Bailey et al. (2020) **JUE**

Micro-geography of individual social capital in cities correlate with the income of residence?

Data


Sándor Juhász
@sandor_juhasz
Can we explain how relatedness of technologies evolves over time? Our new paper in Papers in Regional Science with @TBroekel and @rboschma identifies the drivers behind the widely used concept of #relatedness rsaiconnect.onlinelibrary.wiley.com/doi/full/10.11... Quick thread 1/3
Papers in Regional Science
Explaining the dynamics of relatedness: The role of co-lo...
Relatedness has become a key concept for studying the diversification of firms, regions and countries. However, ...
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1:42 PM · Sep 1, 2020 · Twitter Web App
View Tweet activity


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Researcher at NETI Lab and ANET Lab | Interested in data magic, networks, innovation and (economic) geography
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Joined August 2017
345 Following
211 Followers
Tweets
Tweets & replies
Media
Likes

Data

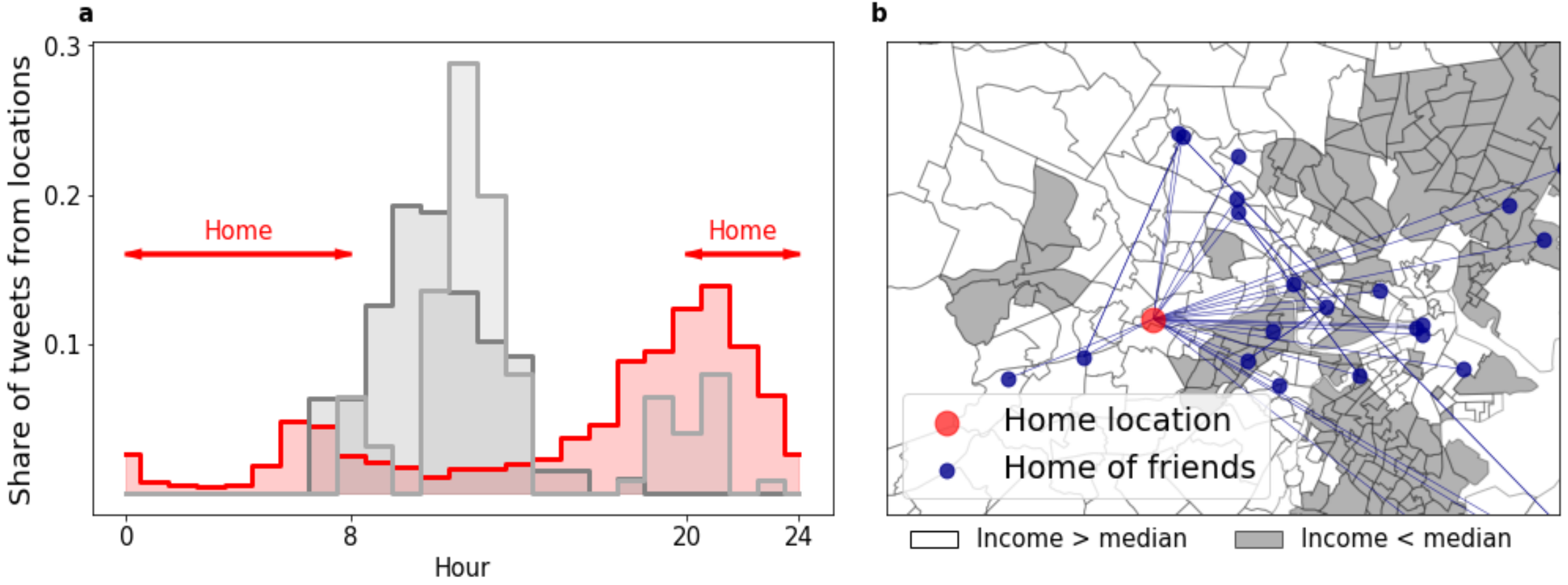


Geolocation possibility

1% of all tweets and 200 tweets of a selected user available every day

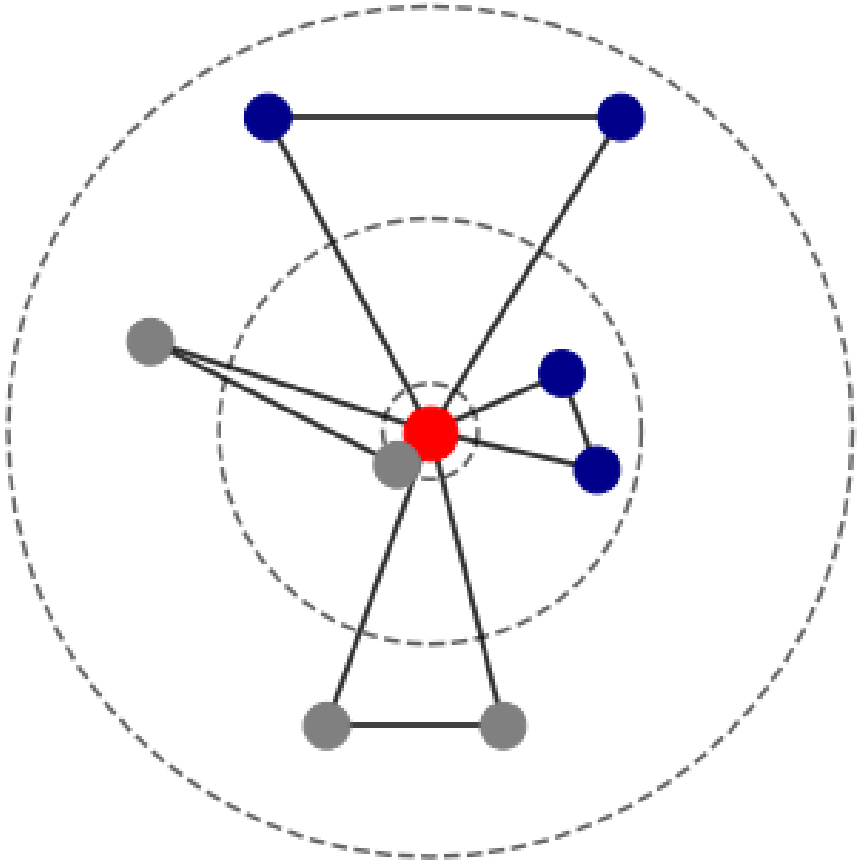
Focus on users with frequent geolocated tweets

Spatial social network of Twitter users

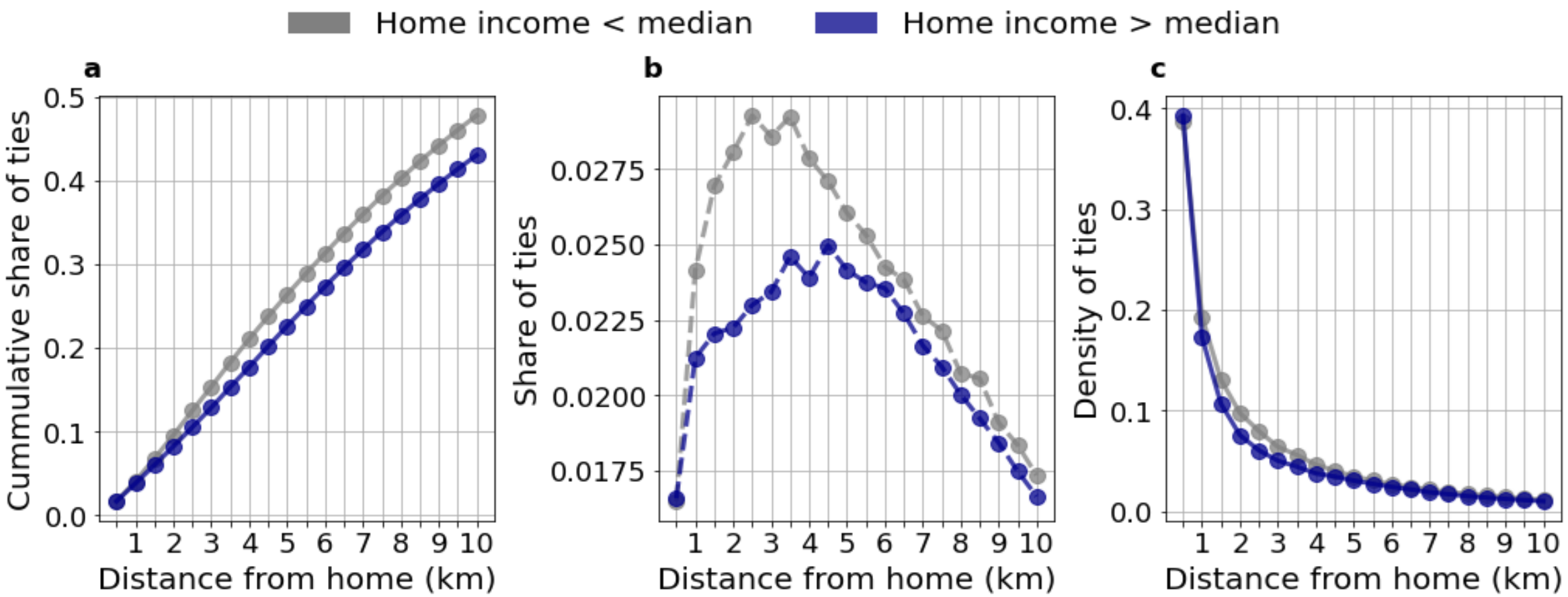


Key network measures

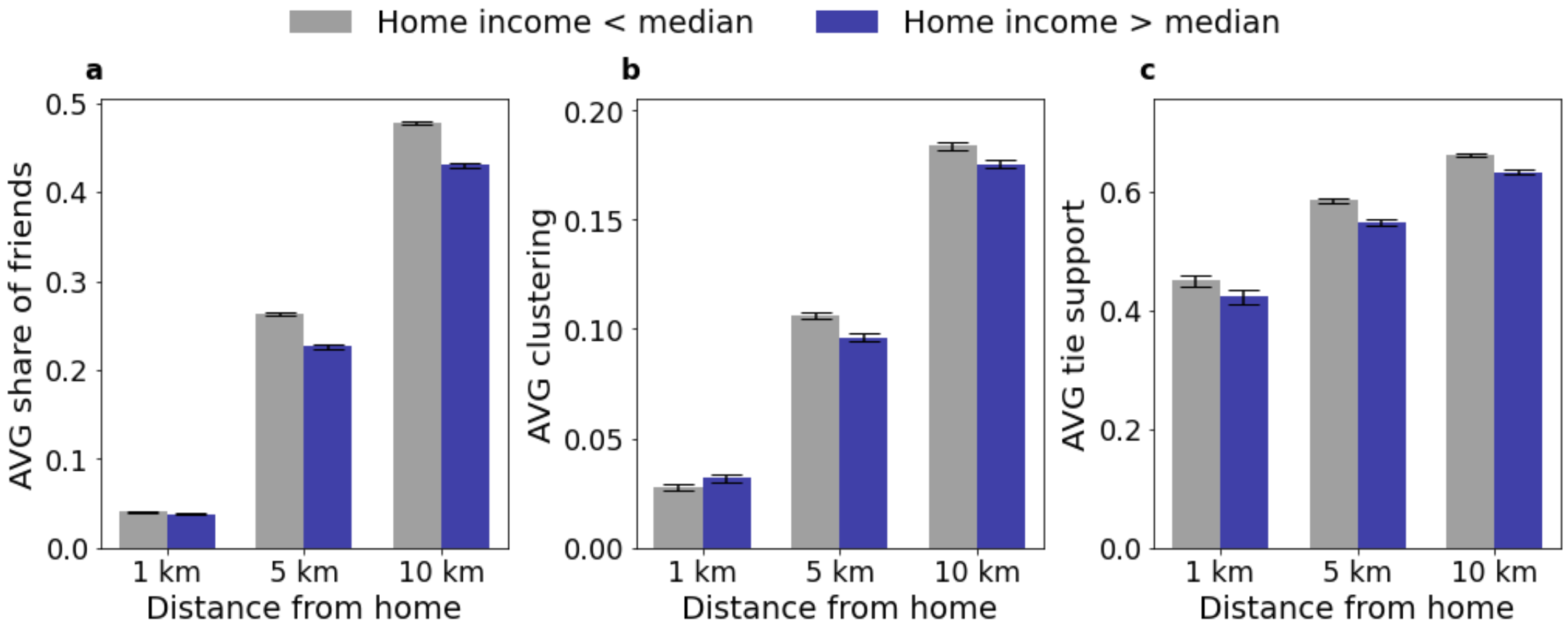
	Full graph	Income < median	Income > median
Degree in 10 km	8	4	4
Share of ties in 1 km	0.125	0.25	0
Share of ties in 5 km	0.375	0.25	0.5
Share of ties in 10 km	1	1	1
Clustering in 1 km	-	-	-
Clustering in 5 km	0.333	-	1
Clustering in 10 km	0.143	0.6	0.6
Support in 1 km	-	-	-
Support in 5 km	0.667	-	1
Support in 10 km	1	1	1

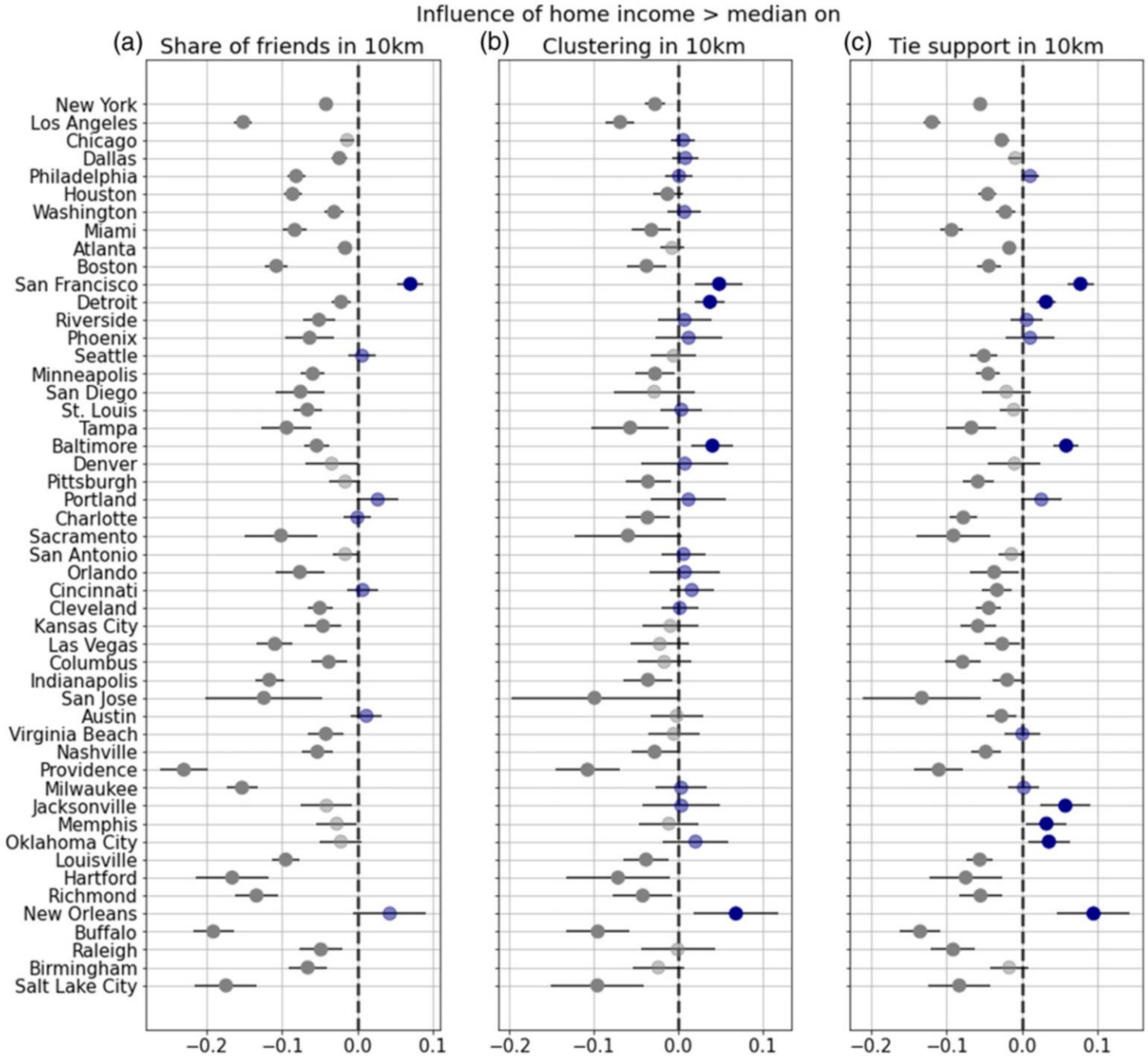


Concentration of social ties around home



Concentration of social ties around home




Influence of home income > median on
(a) Share of friends in 10km
(b) Clustering in 10km
(c) Tie support in 10km
New York
Los Angeles
Chicago
Dallas
Philadelphia
Houston
Washington
Miami
Atlanta
Boston
San Francisco
Detroit
Riverside
Phoenix
Seattle
Minneapolis
San Diego
St. Louis
Tampa
Baltimore
Denver
Pittsburgh
Portland
Charlotte
Sacramento
San Antonio
Orlando
Cincinnati
Cleveland
Kansas City
Las Vegas
Columbus
Indianapolis
San Jose
Austin
Virginia Beach
Nashville
Providence
Milwaukee
Jacksonville
Memphis
Oklahoma City
Louisville
Hartford
Richmond
New Orleans
Buffalo
Raleigh
Birmingham
Salt Lake City
−0.2
−0.1
0.0
0.1

Homophily of connections within 10 km



3.2) Universal patterns of long-distance commuting and social assortativity in cities

Home/work and spatial social network from Twitter messages

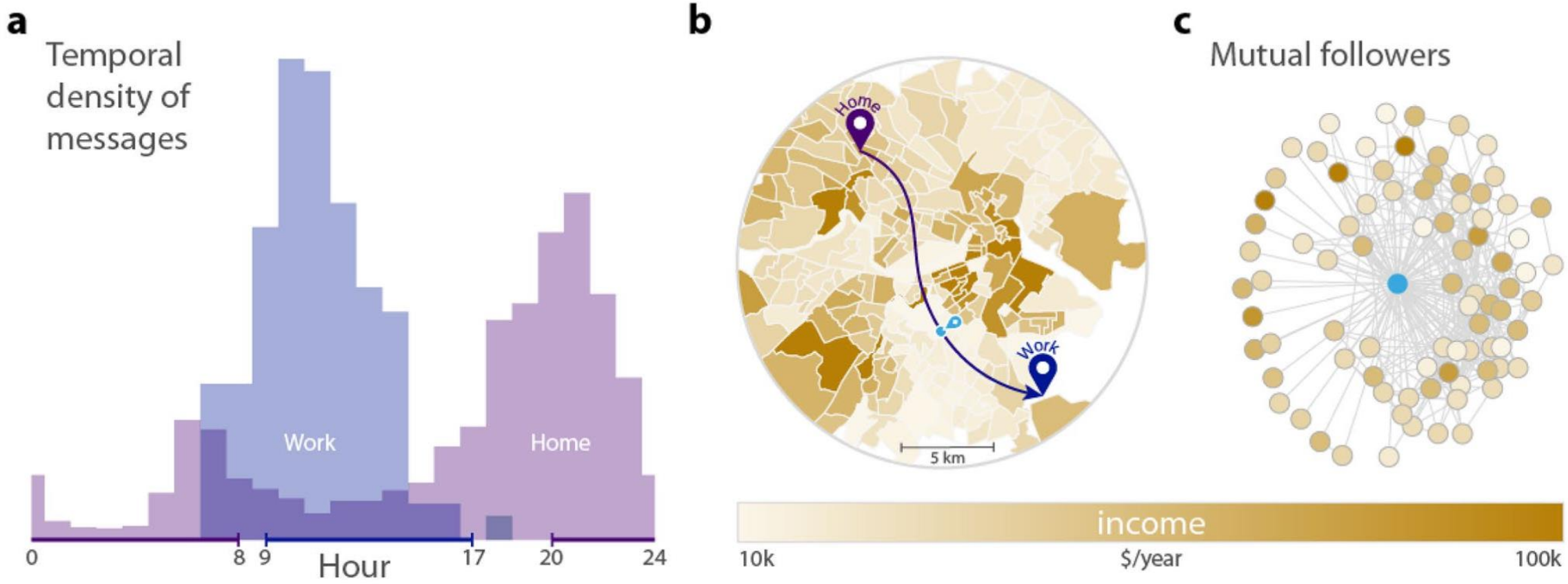
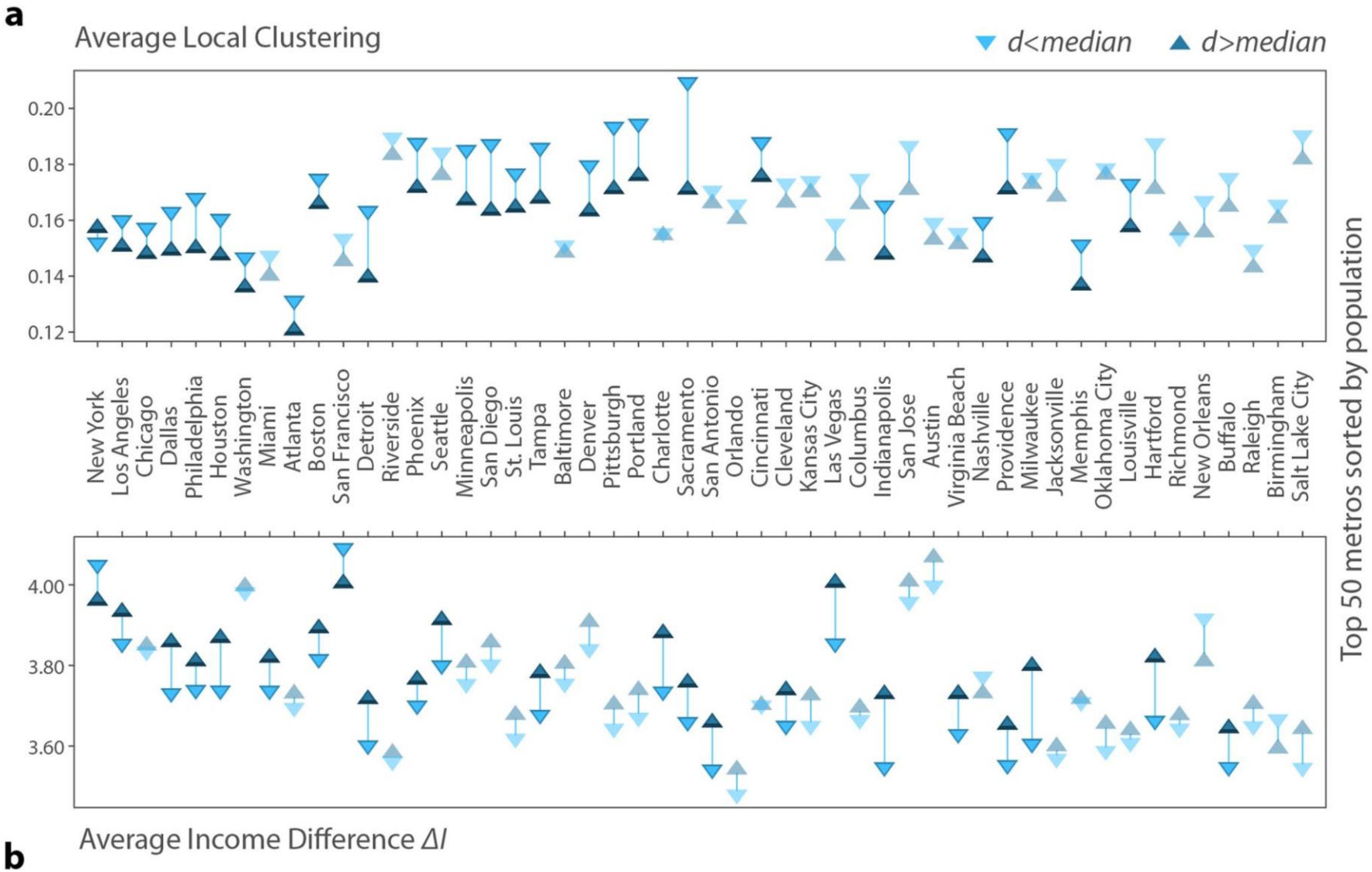
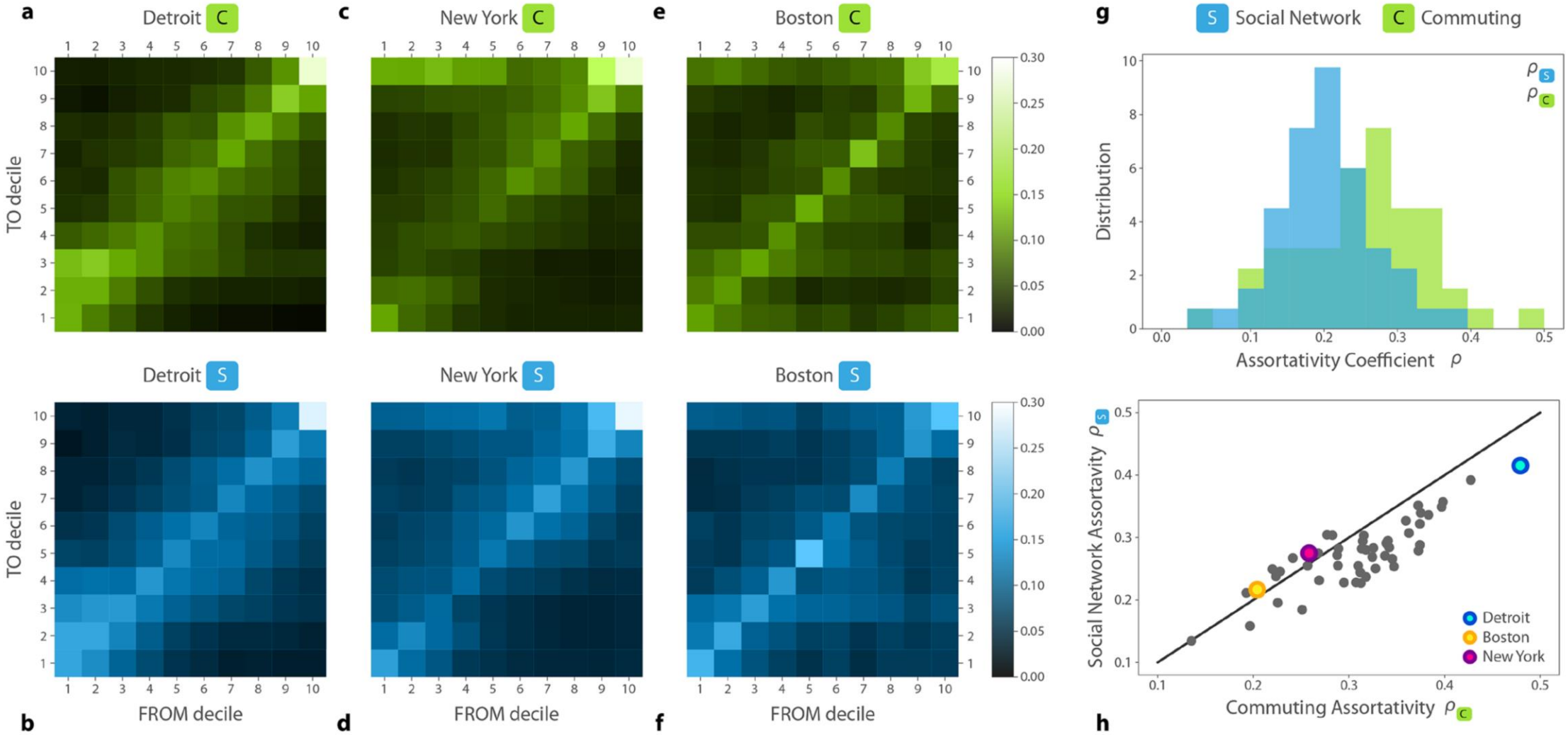


Figure 1. Combination of spatial, temporal and social network data of geolocated Twitter messages. (**a**) Home

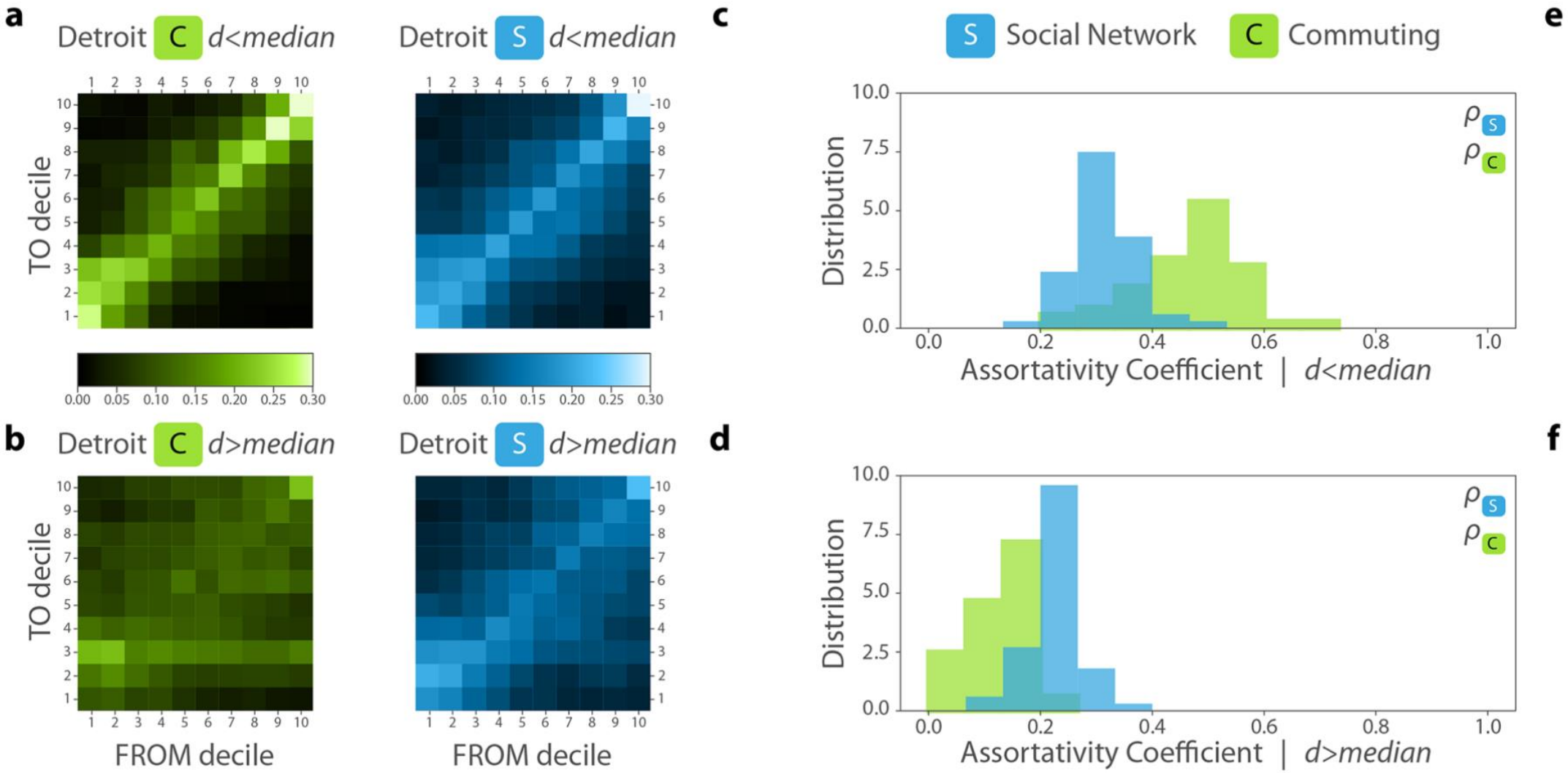
More movement, more open social network



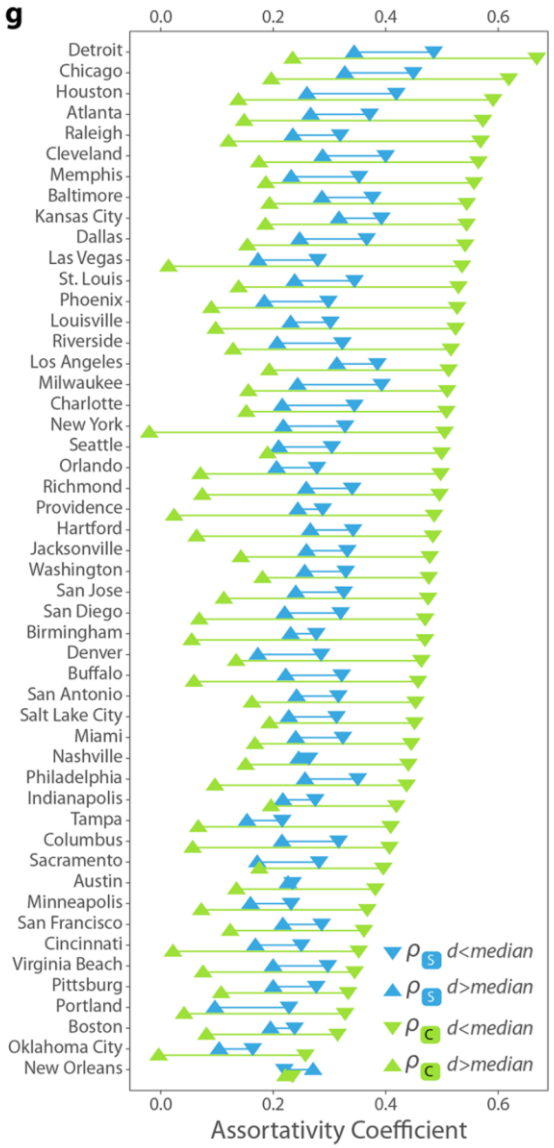
Assortativity in commuting and social relationships

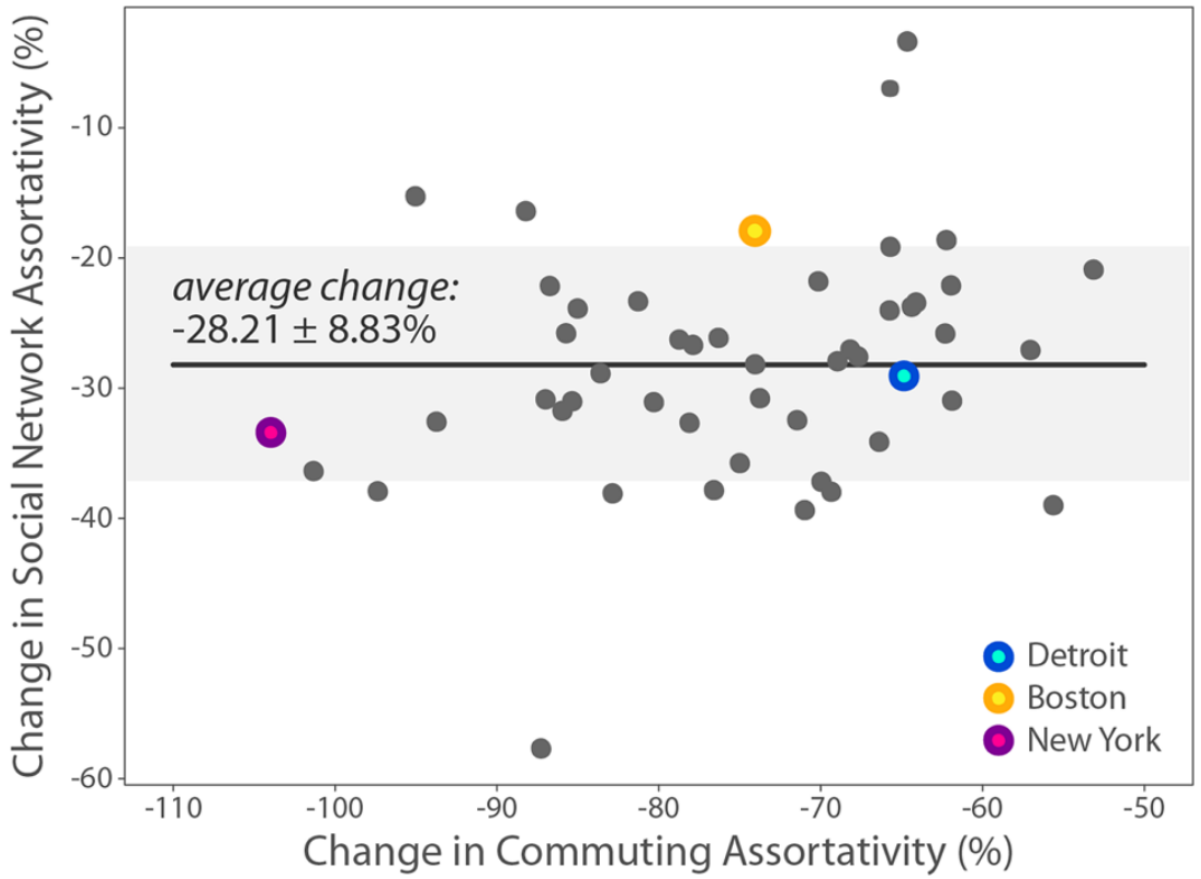


Assortativity in commuting and social relationships



More commuting, lower assortativity of social ties



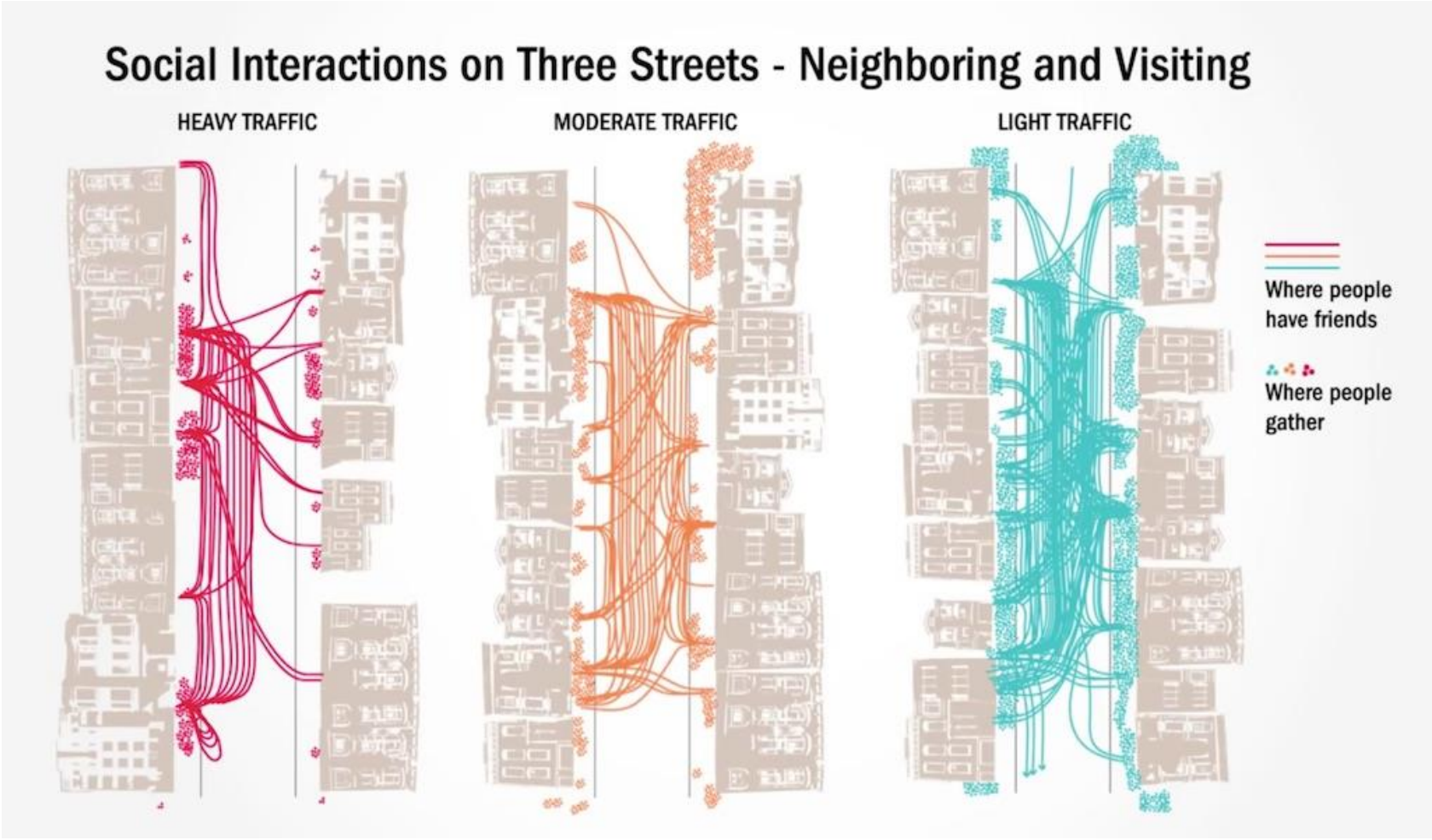


3.3) Urban highways are barriers to social ties

Connecting and dividing at the same time



Highways and social ties



Appleyard, D. (1981) Livable Streets

Highways and social ties

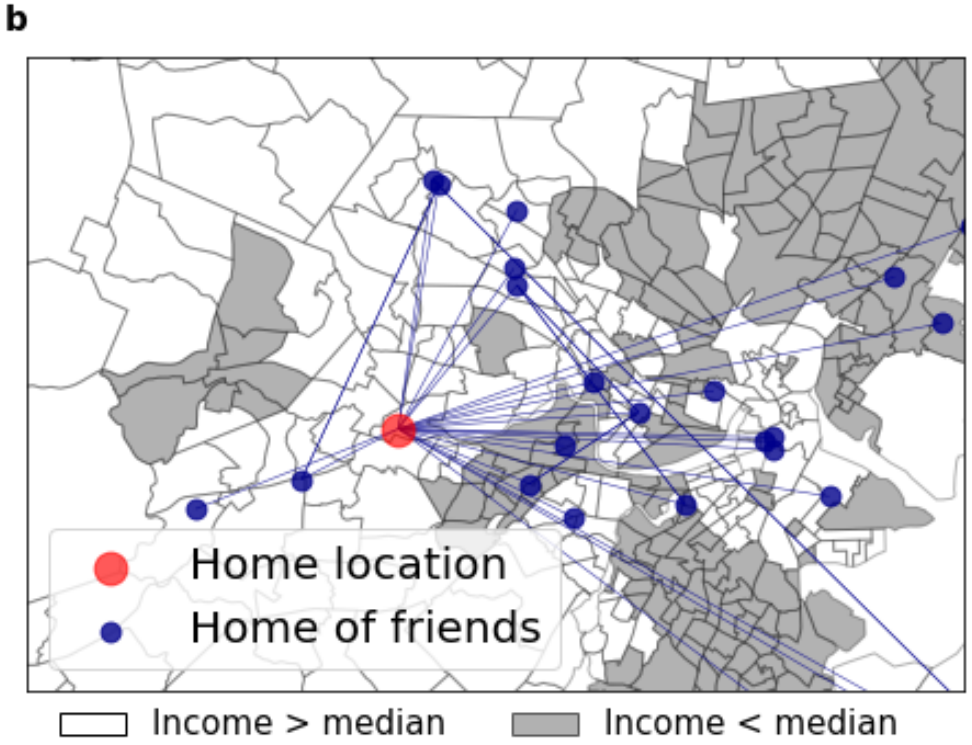
Large-scale quantitative evidence on how highways interact with social ties within cities



https://www.theguardian.com/cities/2018/feb/21/roads-nowhere-infrastructure-american-inequality

Social ties

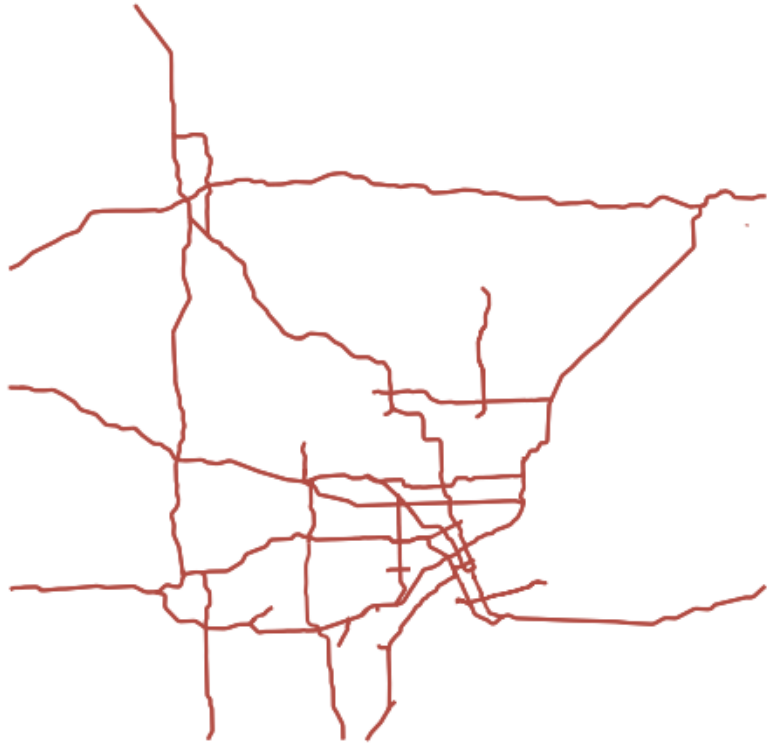
Twitter user activity (2012-2013)
1M users, 2.7M connections
in the top 50 US metropolitan areas



Kovács et al. (2022) **EPB**

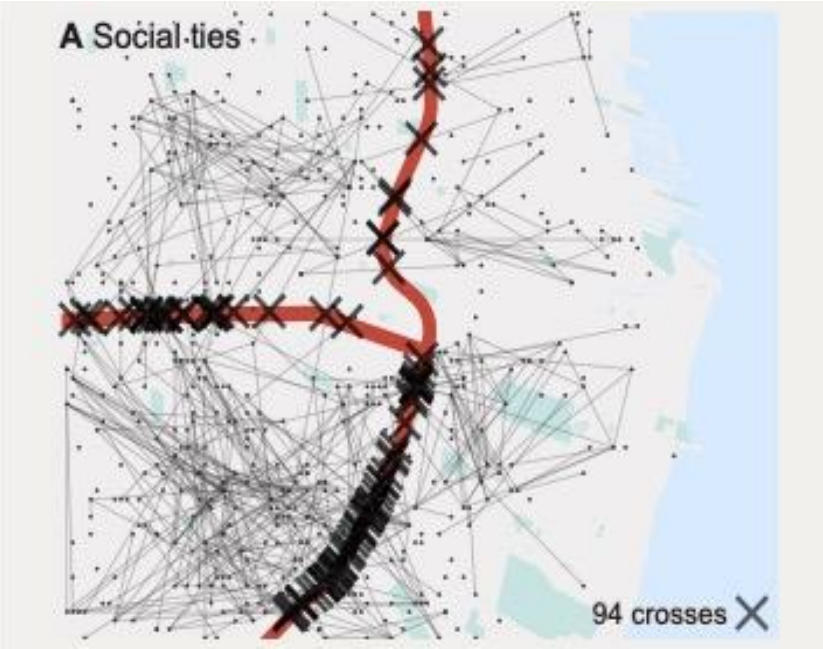
Highways

OpenStreetMap (OSM)
(*motorway* + *trunk links*)

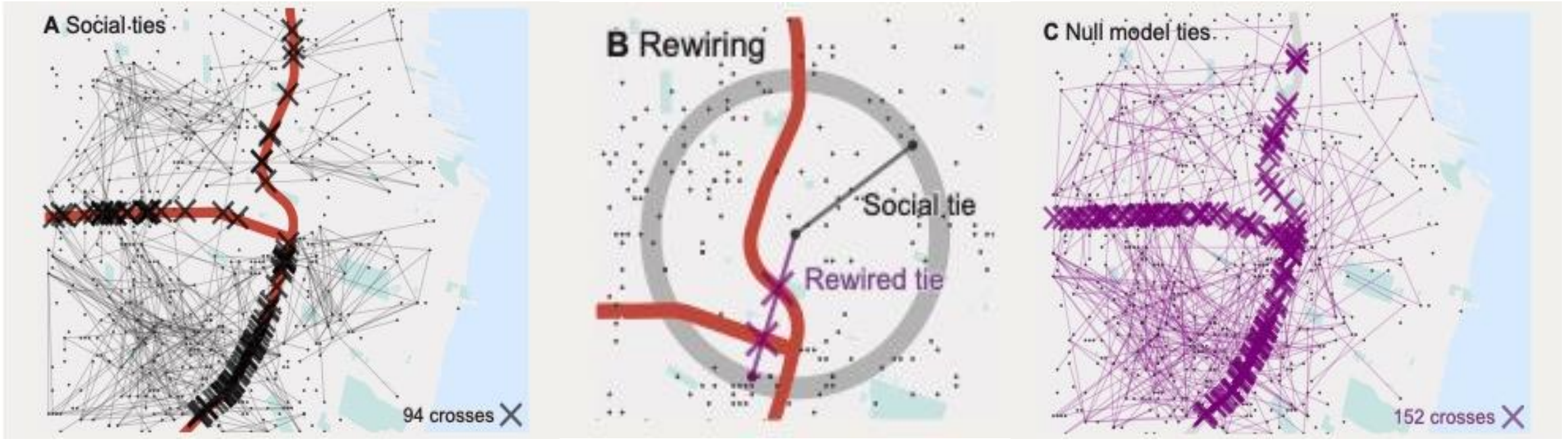


Detroit (-84.158, 42.028, -82.334, 43.327)

Barrier Score

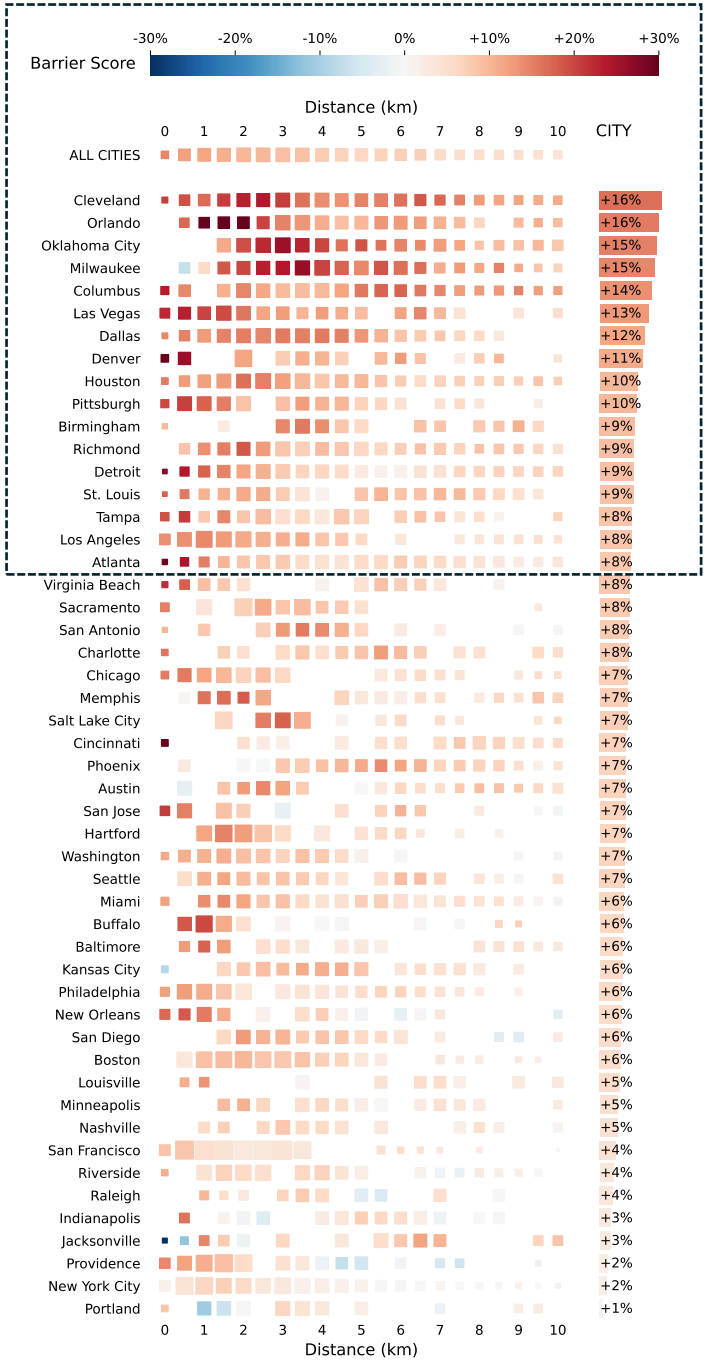


Barrier Score



D Barrier Score

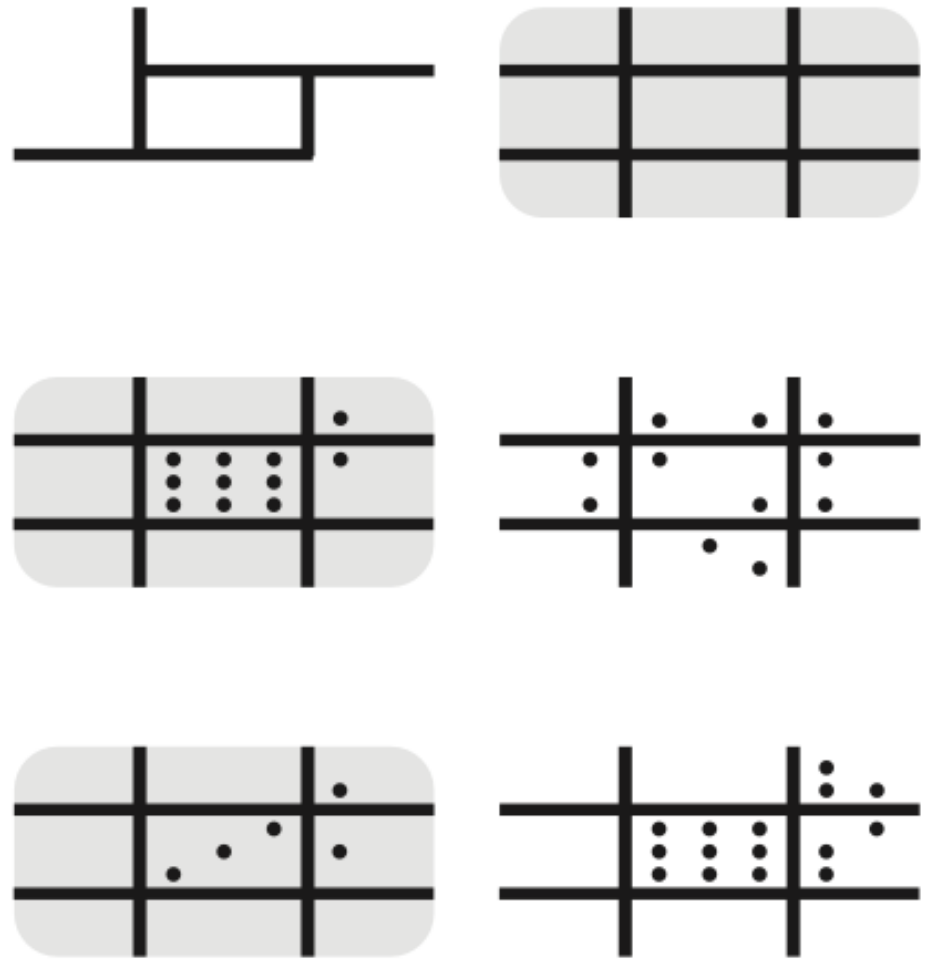
$$B_i = \frac{152\text{-}94}{94} = \mathbf{+62\%}$$



-30% -20% -10% 0% +10% +20% +30%
Barrier Score
Distance (km)
0 1 2 3 4 5 6 7 8 9 10
CITY
ALL CITIES
Cleveland +16%
Orlando +16%
Oklahoma City +15%
Milwaukee +15%
Columbus +14%
Las Vegas +13%
Dallas +12%
Denver +11%
Houston +10%
Pittsburgh +10%
Birmingham +9%
Richmond +9%
Detroit +9%
St. Louis +9%
Tampa +8%
Los Angeles +8%
Atlanta +8%

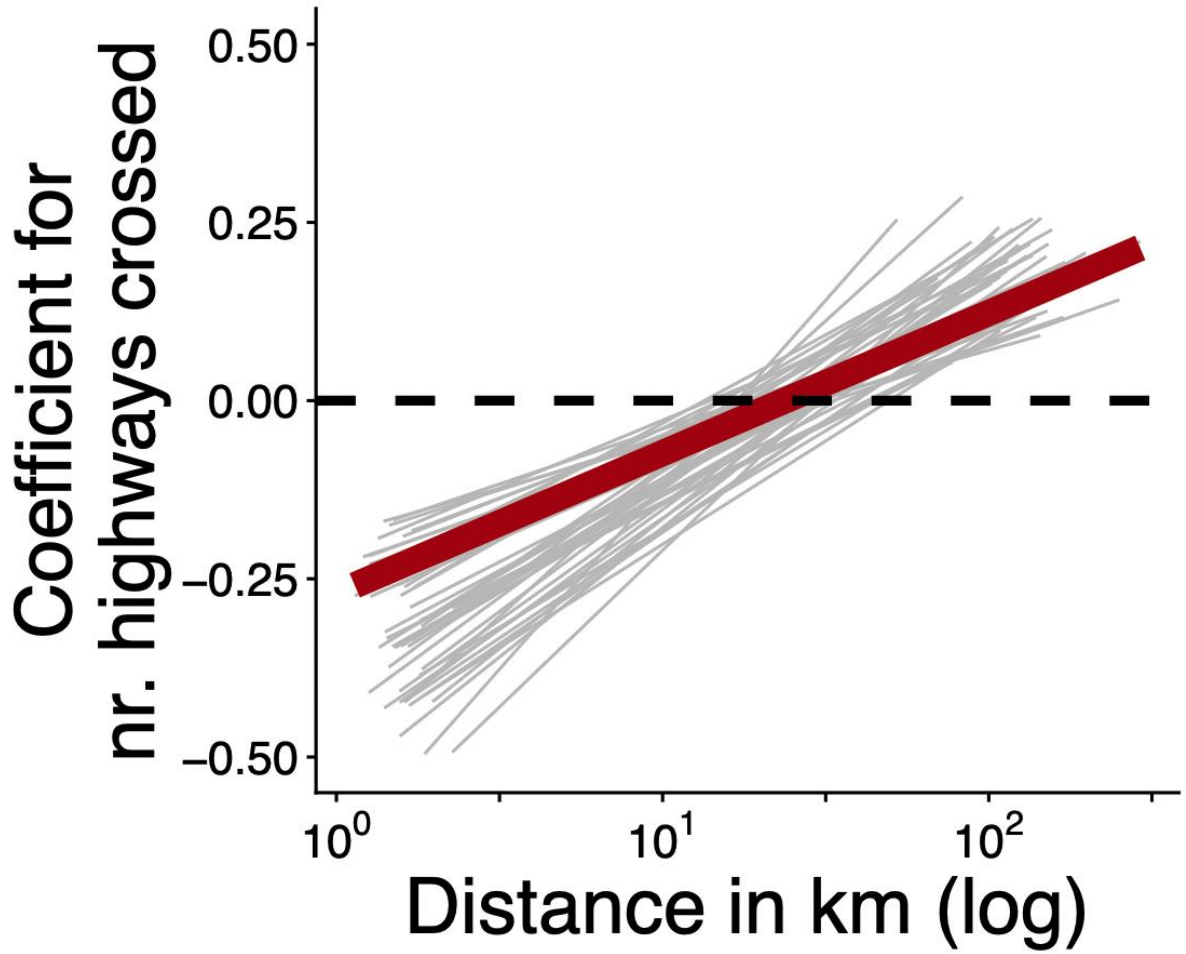
Barrier Score

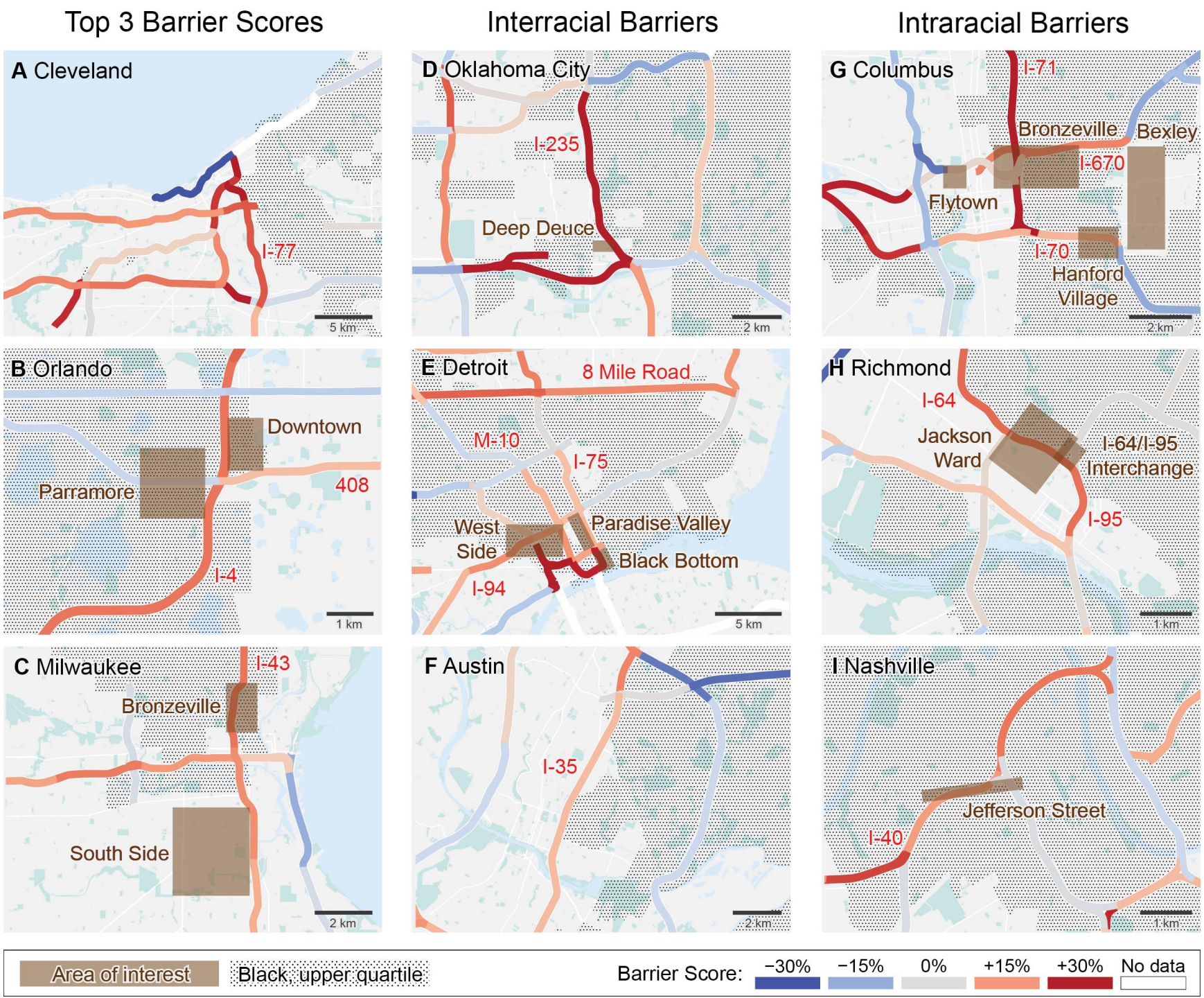
Highway length (log)	0.469*** (0.139)
Fragmentation (log)	−0.257** (0.129)
User population density	−0.390*** (0.139)
Constant	0.000 (0.125)



	Number of social ties (log)					
	(1)	(2)	(3)	(4)	(5)	(6)
Nr. of highways crossed (log)		−0.025***				−0.013***
		(0.003)				(0.003)
Nr. of railways and waterways crossed (log)			−0.018***			−0.013***
			(0.005)			(0.004)
Income abs. difference				−0.019***		−0.018***
				(0.002)		(0.001)
Racial similarity					0.026***	0.023***
					(0.003)	(0.003)
Distance (log)	−0.101***	−0.086***	−0.087***	−0.099***	−0.101***	−0.081***
	(0.013)	(0.014)	(0.016)	(0.013)	(0.013)	(0.015)
User population (product log)	0.029***	0.027***	0.026***	0.026***	0.024***	0.019**
	(0.008)	(0.007)	(0.008)	(0.008)	(0.008)	(0.008)
Metro fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,669,688	2,669,688	2,669,688	2,669,688	2,669,688	2,669,688
R^2	0.042	0.043	0.043	0.045	0.047	0.051

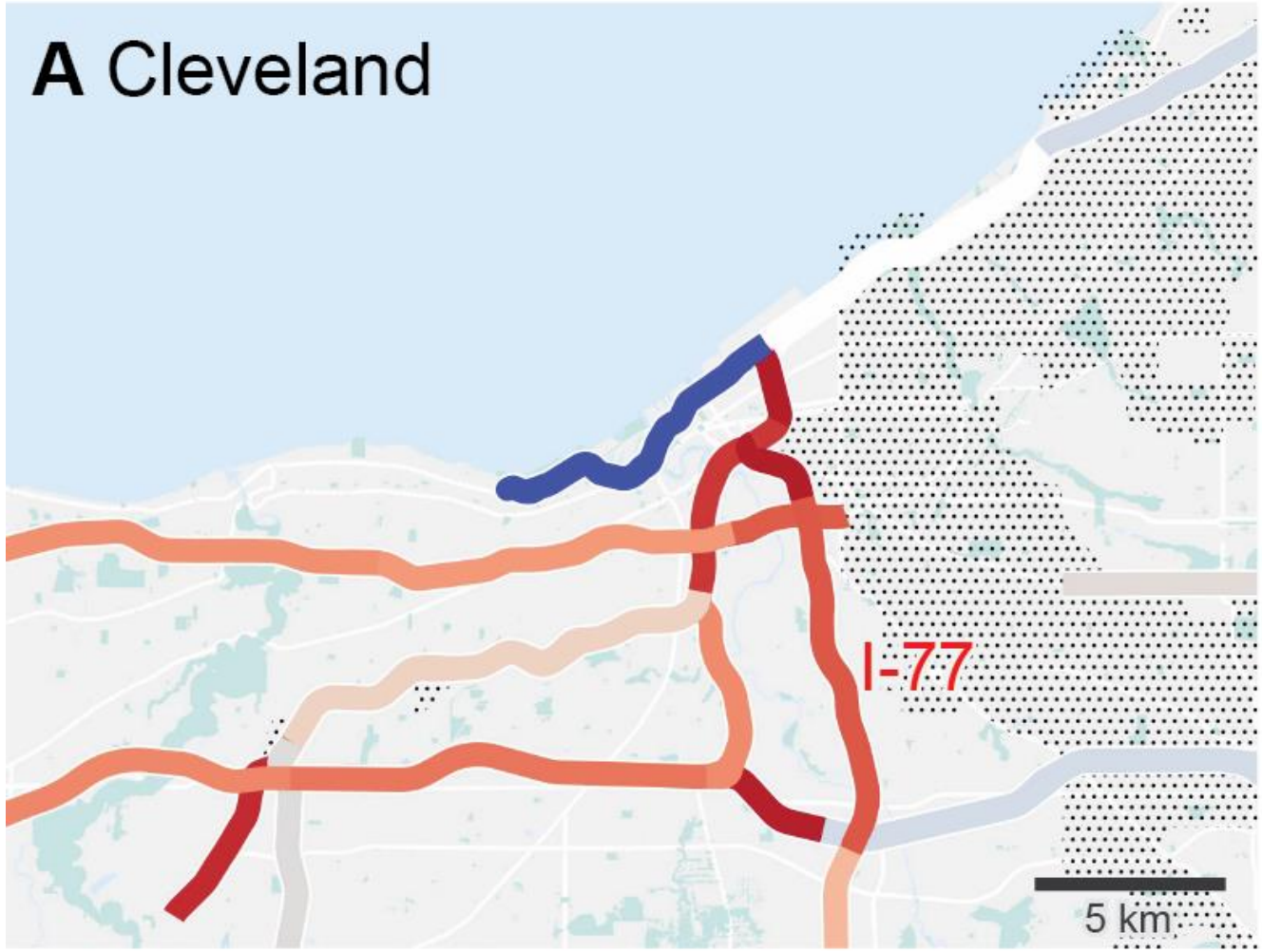
Estimator	OLS	OLS
Dependent variable	$log(1+t_{ij})$	$log(1+t_{ij})$
	(1)	(2)
Nr highways crossed (log)	−0.013***	−0.270***
	(0.003)	(0.013)
Distance (log)	−0.081***	−0.238***
	(0.015)	(0.013)
Distance X Nr highways crossed		0.197***
		(0.008)
Nr. of railways and waterways crossed (log)	−0.013***	−0.013***
	(0.003)	(0.004)
Income abs difference	−0.018***	−0.016***
	(0.001)	(0.001)
Racial similarity	0.023***	0.021***
	(0.003)	(0.003)
Population (product log)	0.019**	0.018***
	(0.008)	(0.007)
Metro fixed effect	Yes	Yes
Observations	2,669,688	2,669,688
R^2	0.051	0.064




Top 3 Barrier Scores
Interracial Barriers
Intraracial Barriers
A Cleveland
I-77
5 km
D Oklahoma City
I-235
Deep Deuce
2 km
G Columbus
I-71
Bronzeville
Bexley
I-670
Flytown
I-70
Hanford Village
2 km
B Orlando
Downtown
Parramore
408
I-4
1 km
E Detroit
8 Mile Road
M-10
I-75
West Side
Paradise Valley
Black Bottom
I-94
5 km
H Richmond
I-64
Jackson Ward
I-64/I-95 Interchange
I-95
1 km
C Milwaukee
I-43
Bronzeville
South Side
2 km
F Austin
I-35
2 km
I Nashville
Jefferson Street
I-40
1 km
Area of interest
Black, upper quartile
Barrier Score:
−30%
−15%
0%
+15%
+30%
No data

High Barrier Scores

I-77 separates the cluster of predominantly Black communities in the east from the rest of the metropolitan area

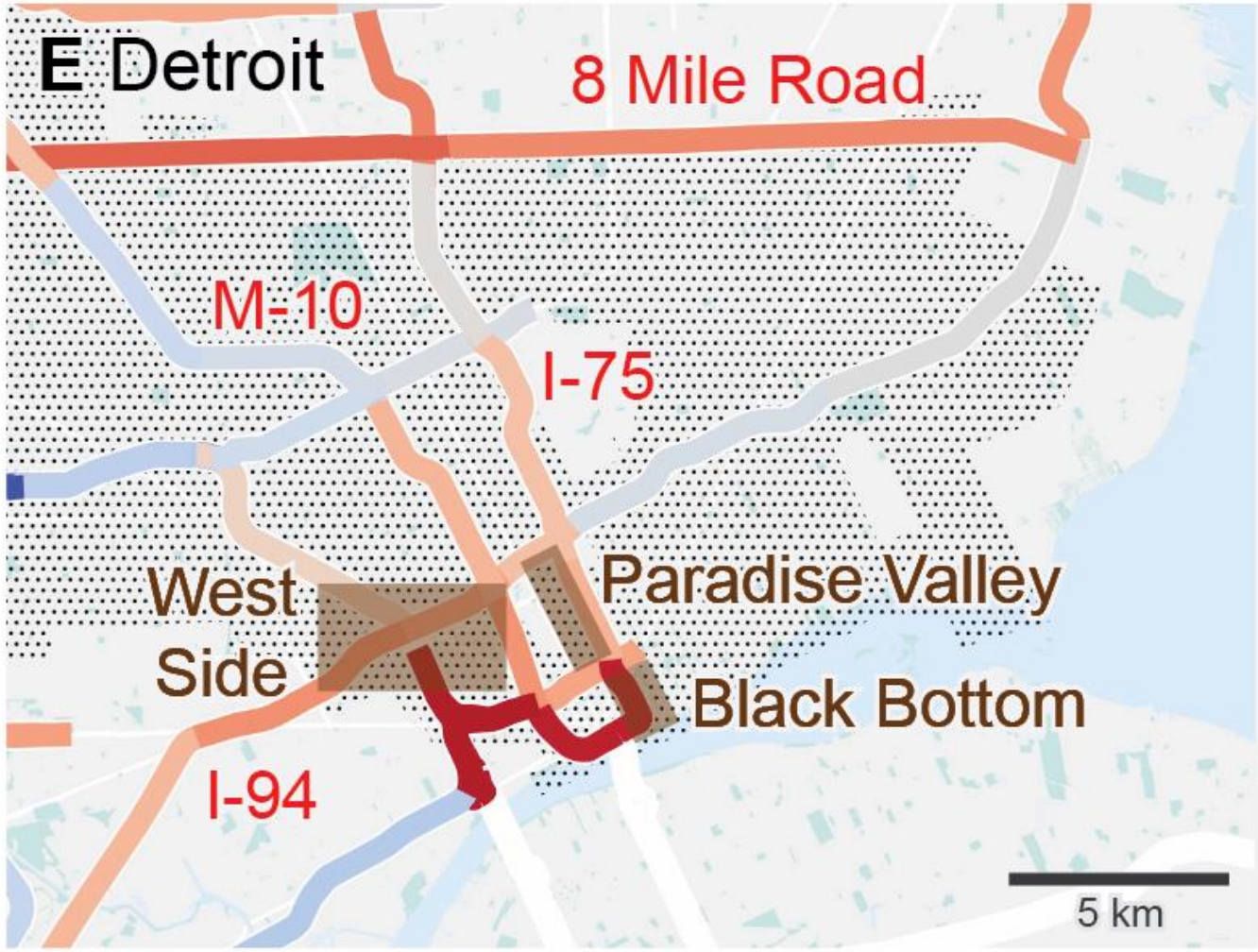


Area of interest | Black, upper quartile | Barrier Score: −30% | −15% | 0% | +15% | +30% | No data

Interracial Barriers

The Birwood Wall (Eight Mile Wall) manifesting the divide between the Black inner city and the white suburbs

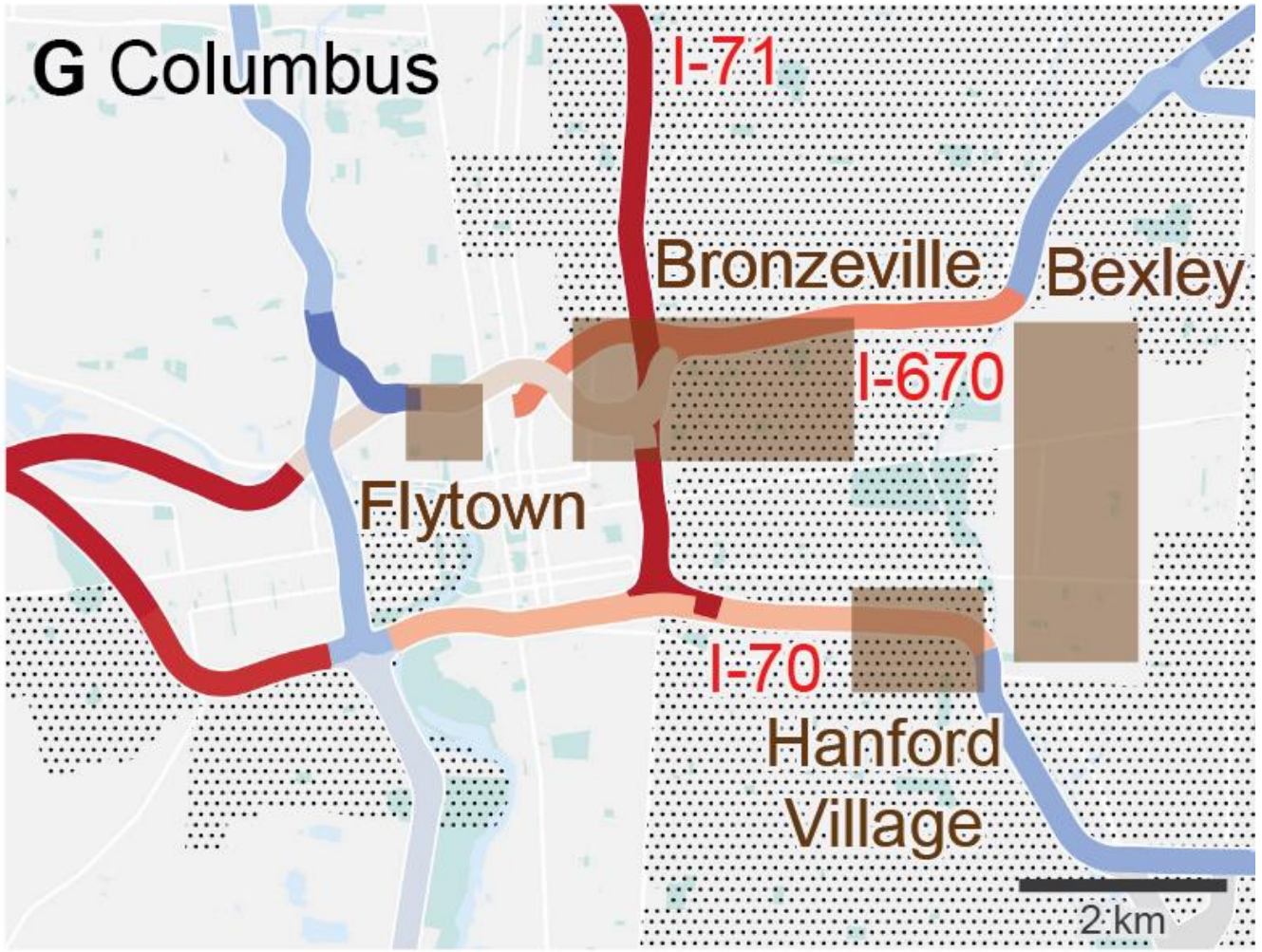
Very high Barrier Scores in the inner city close to the historically Black neighborhoods



Area of interest | Black, upper quartile | Barrier Score: −30% | −15% | 0% | +15% | +30% | No data

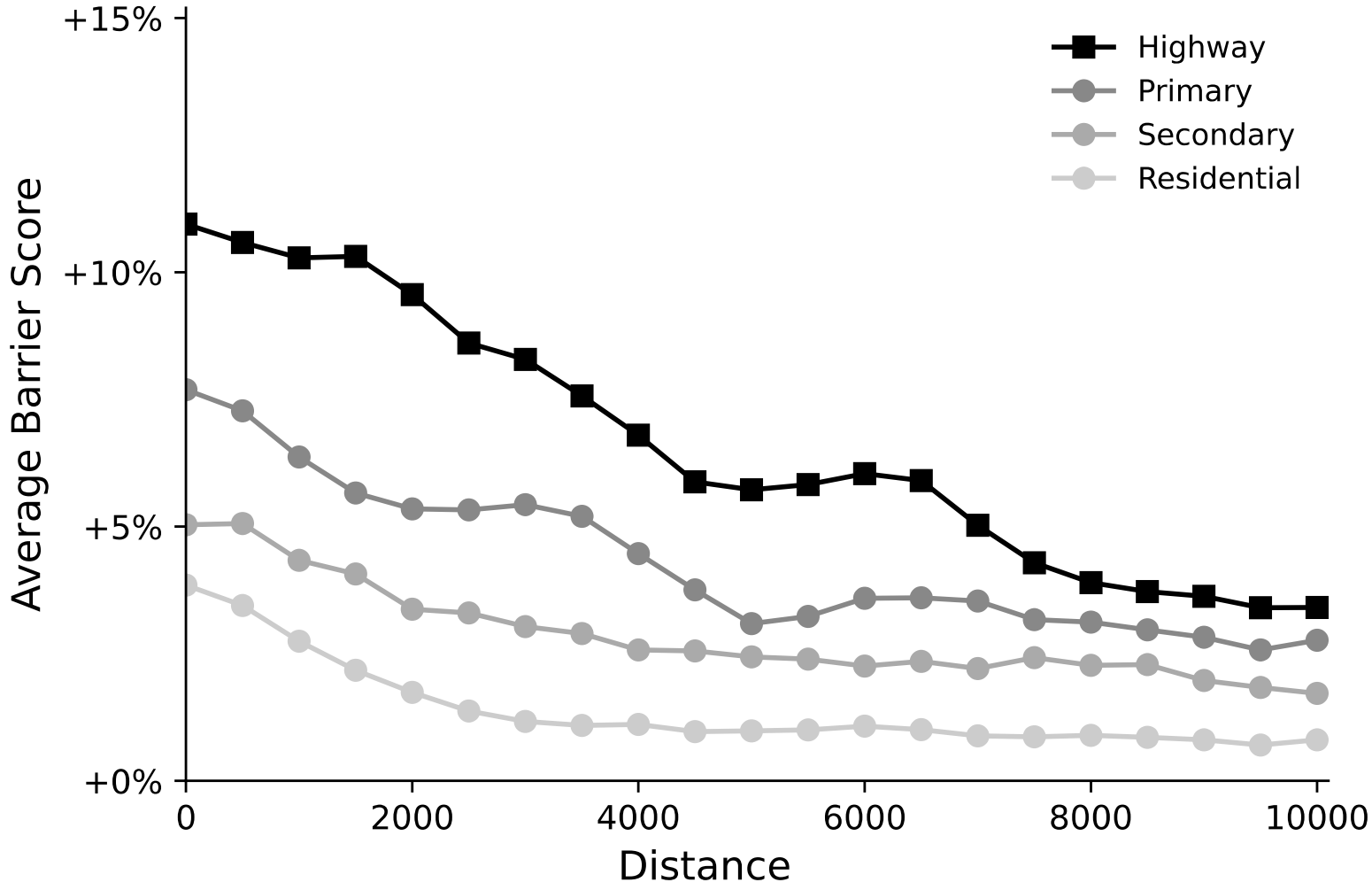
Intraracial Barriers

Highway construction as deliberate destruction of economically disadvantaged and predominantly Black neighborhoods

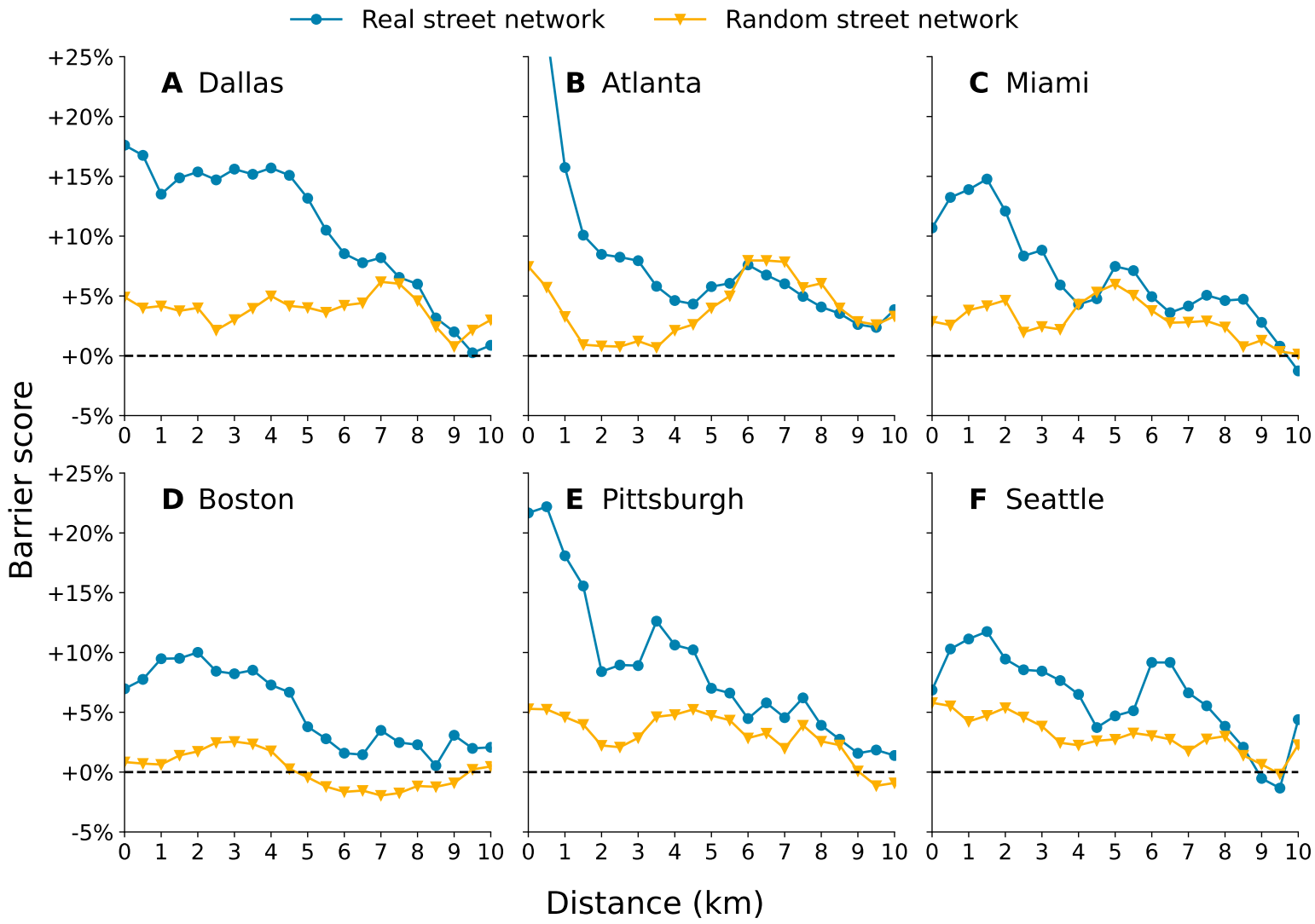


Area of interest | Black, upper quartile | Barrier Score: −30% | −15% | 0% | +15% | +30% | No data

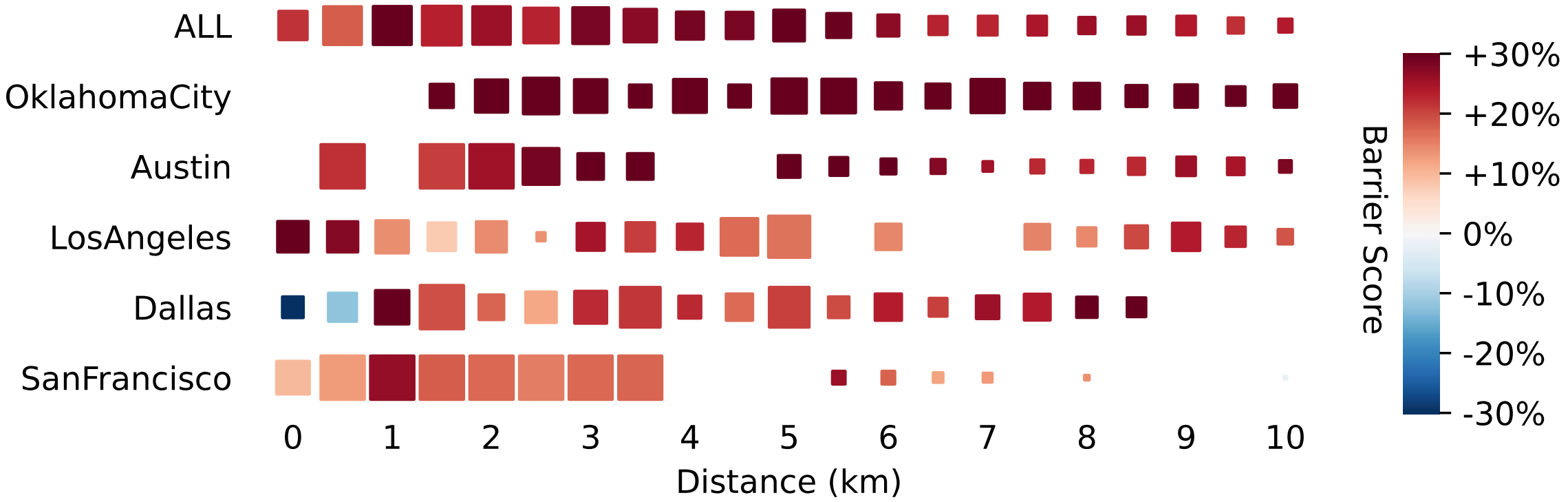
Robustness checks – other road categories



Robustness checks – random highway network



Robustness checks – Gowalla data



Discussion

Adding a highly granular perspective, we quantify the intuition that urban highways are barriers to social ties

Highways facilitate connections over longer distances, but at considerable harms with long lasting effects

Enrich the debate on contemporary highway policies and contribute to a more careful, evidence-based urban planning

Thank you!

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